



Set-Up Instructions | PUBLIC

SAP S/4HANA Cloud Public Edition

2026-06-16

Setting Up *Invoice Processing with SAP Ariba Invoicing* (4N6)

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1 Setup Overview

📘 Note

Starting from June 2024, we adjusted the structure of this set-up guide to highlight the usage of booster.

The booster does not apply to customers in the following cases. Follow the 2402 version of 4N6 setup instruction guide in [3472212](#) to complete your system setup manually if:

- You have already started the setup process, for example, have finished the SAP BTP and integration setup.
- Your SAP S/4HANA Cloud Public Edition already has an integration with SAP Master Data Integration.
- Your SAP S/4HANA Cloud Public Edition is connected with a system like SAP SuccessFactors, which already uses SAP Master Data Integration service.

📘 Note

We recommend using the Cloud Integration Automation Service (CIAS) tool to set up this integration scenario instead of using manual configuration. CIAS provides a guided workflow that features:

- Customer-specific system information (for example, target URLs)
- Reusable configuration settings between tasks
- Automated execution capabilities

With the included project management tool and complementary support information, CIAS helps reduce the time you need for manual tasks. For more information, see the [product page](#) on the SAP Help Portal.

To access CIAS, please choose one of the following options:

- 1. [Subscription to Standard Plan](#)
- 2. [Assigning Roles to the Users](#)
- 3. [Using Cloud Integration Automation Service Application](#)

📘 Note

You can find the scenario from under tab ► *Integration Solutions and Scenarios* ► *SAP S/4HANA Cloud Public Edition Integration* ►.

The scenario name you are looking for is **SAP Ariba Invoicing (4N6)**.

Alternatively, you can also enter the key word "4N6" in the search box to reduce the number of solutions shown under *Integration Solutions and Scenarios* tab and quickly get to scenario 4N6.

- [Navigate directly to the Maintenance Planner to start the guided CIAS workflow via this link.](#)

SAP Ariba Invoicing on SAP Business Technology (BTP) provides a unified solution for receiving and managing supplier invoices, with a connection to different backend systems like SAP S/4HANA Cloud Public Edition.

To release its full function, SAP Ariba Invoicing requires multiple tenants, for example test tenant 1, test tenant 2, and production tenant 1. The general setup steps for each SAP Ariba Invoicing tenant are listed below:

Table 1: Setup Steps

Description	Chapter
Understand SAP Ariba Invoicing system landscape	System Landscape for SAP Ariba Invoicing [page 6] System Landscape with Test, Demo and Development (TDD) License [page 7]
Activate scope item 4N6	Activate Scope Item Invoice Processing with SAP Ariba Invoicing (4N6) and Data Protection and Privacy (5LE) [page 9]
Run the booster in SAP BTP cockpit	Set up Configuration Using Booster in SAP BTP Cockpit [page 11]
Master Data Integration via SAP Integration Suite, managed gateway	Setup SAP Integration Suite, managed gateway [page 14]
Configure the Connected System - SAP S/4HANA Cloud Public Edition	
Create communication user	Create Communication User [page 17]
Create communication system	Create Communication System [page 17]
Set up Accounting Master Data Integration (SAP_COM_0179)	Set Up Accounting Master Data Integration (SAP_COM_0179) [page 19]
Configure business name and SOAP replication for business partner	Configure Business System Name and Business Partner SOAP Replication [page 52]
Add SAP Ariba Invoicing to your system HTTP allowlist	Add SAP Ariba Invoicing to your system HTTP Allowlist [page 23]
Set up target system for user replication	Set up Target System for User Replication [page 89]
Deactivate default identity provider user logon	Deactivate Default Identity Provider User Logon [page 25]
Complete setup with the steps described in the SAP Ariba Invoicing Administration Guide	Further Operations [page 28]

2 System Landscape

2.1 System Landscape for SAP Ariba Invoicing

In this chapter we describe the recommended system landscape for 3 SAP Ariba Invoicing tenants. You can also refer to the system landscape setup image below for more information.

This system landscape description applies to all SAP Ariba Invoicing solutions except that of the *test*, *demo* and *development* license.

Test Tenant

1. The first SAP Ariba Invoicing **test** tenant should be connected to SAP S/4HANA Cloud Public Edition **test** tenant.
2. The second SAP Ariba Invoicing **test** tenant should be connected to SAP S/4HANA Cloud Public Edition **project line test** tenant when it is needed.

Production Tenant

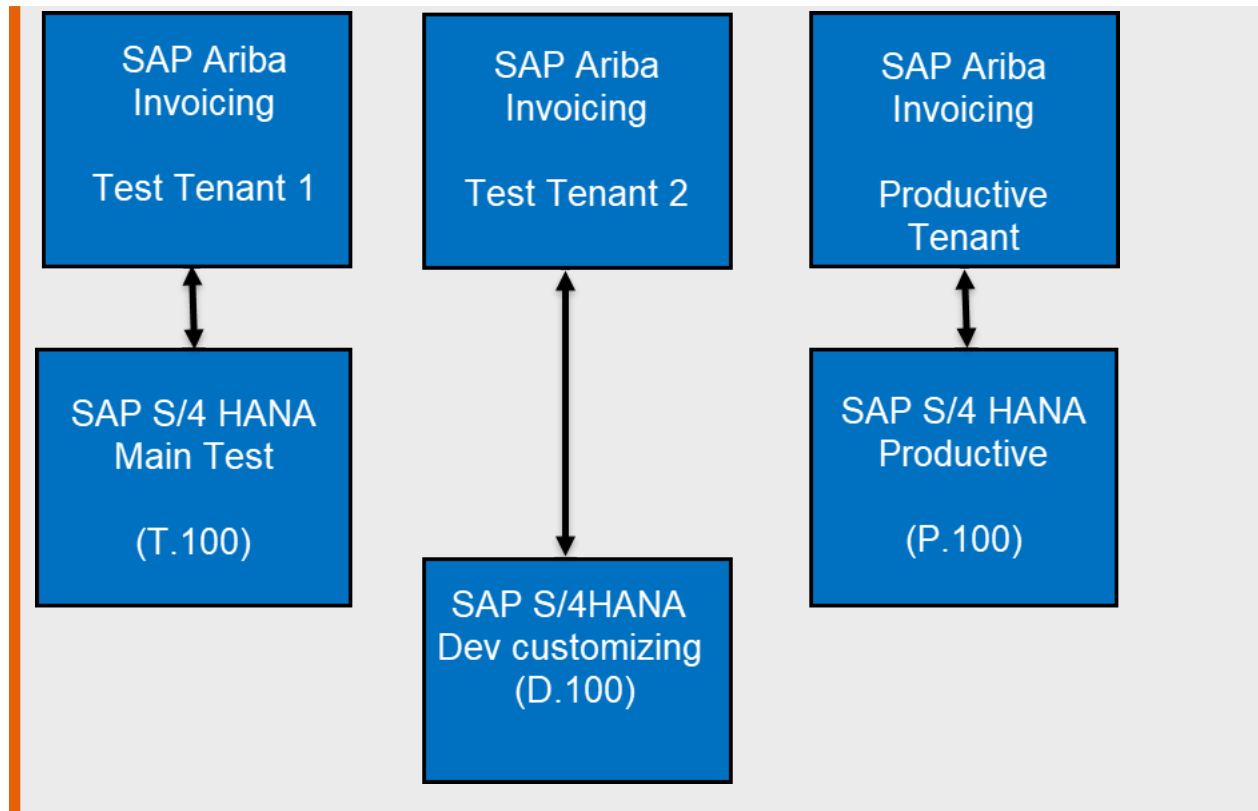
1. SAP Ariba Invoicing **production** tenant could only be connected to SAP S/4HANA Cloud Public Edition **production** tenant.

📌 Note

1. It is not allowed to:
 - Connect multiple SAP S/4HANA Cloud Public Edition tenants (for example, development tenant and test tenant) with the same SAP Ariba Invoicing tenant.
 - Connect SAP Ariba Invoicing **test** tenant with SAP S/4HANA Cloud Public Edition **production** system.
 - Connect SAP Ariba Invoicing **production** tenant with SAP S/4HANA Cloud Public Edition non-production systems (**development system**, **test system**, **project line test system** and etc).
2. It is not recommended to:
 - Connect SAP S/4HANA Cloud Public Edition **starter** tenant and SAP S/4HANA Cloud Public Edition **development** tenant with SAP Ariba Invoicing **test** tenant.

⚠ Caution

The system connections could not be reused, or be redirected for a new connection. For example, the connection between SAP Ariba Invoicing **test** tenant and SAP S/4HANA Cloud Public Edition **development** system can not simply be redirected and changed to a connection between SAP Ariba Invoicing **test** tenant and SAP S/4HANA Cloud Public Edition **test** system.



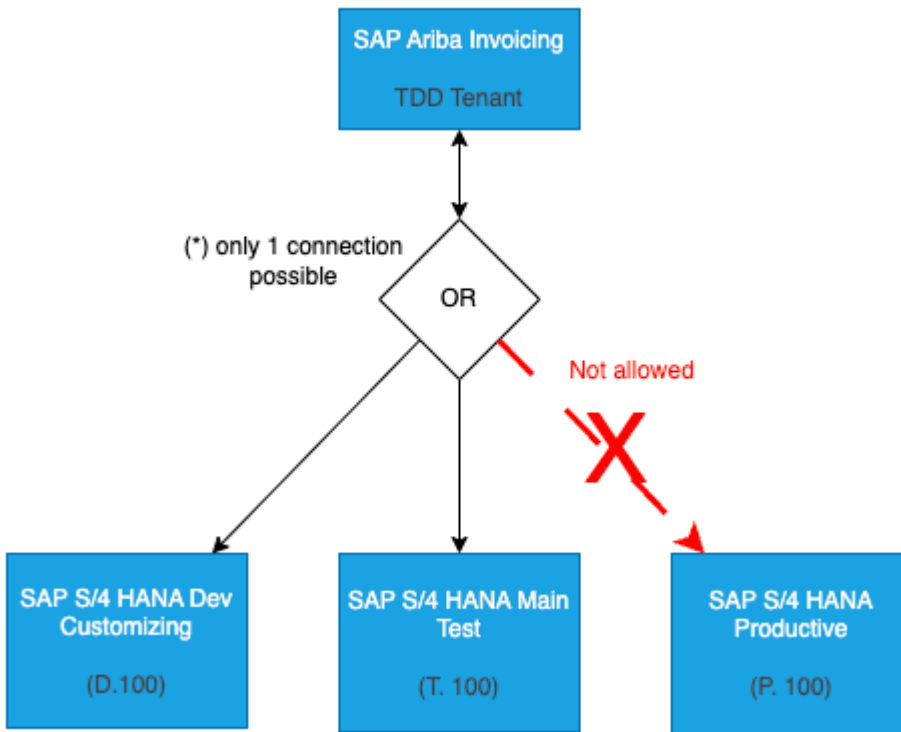
2.2 System Landscape with Test, Demo and Development (TDD) License

The SAP Ariba Invoicing tenant created with *Test, Demo and Development* license can connect with SAP S/4HANA test system, SAP S/4HANA demo system or SAP S/4HANA development system, but it can only connect with one system. Therefore, only **one** of these system tenants should be connected.

This tenant is not allowed to connect with an SAP S/4HANA production system. Please refer to the image below.

⚠ Caution

The system connections could not be reused, or be redirected for a new connection. For example, the connection between SAP Ariba Invoicing **test** tenant and SAP S/4HANA **development** system can not simply be redirected and changed to a connection between SAP Ariba Invoicing **test** tenant and SAP S/4HANA **test** system.



3 **Activate Scope Item Invoice Processing with SAP Ariba Invoicing (4N6) and Data Protection and Privacy (5LE)**

Prerequisite

Before activating 4N6, it is mandatory to check the communication scenario of your SAP S/4HANA Cloud Public Edition system. For the booster to work, your SAP S/4HANA Cloud Public Edition system must not have existing communication scenarios for SAP Master Data Integration and SAP Master Data Orchestration.

Procedure

It is mandatory that the scope item *Invoice Processing with SAP Ariba Invoicing (4N6) and Data Protection and Privacy (5LE)* is active.

Depending on your configuration environment for SAP S/4HANA Cloud Public Edition, choose one of the following options to check the status:

- If you have **SAP Central Business Configuration**:
In the *Scope and Organizational Structure* phase, navigate to the *Activities* tab. Search for *Define Scope* and choose *Open*.
If this scope item is not activated, you can activate it by selecting the scope item during system scoping.
- If you only have the **Manage Your Solution** app in the SAP S/4HANA Cloud Public Edition system:
In the SAP S/4HANA Cloud Public Edition system, in the SAP Fiori launchpad, open the *Manage Your Solution* app and go to *View Solution Scope*.
If the scope item is not active, please request the activation via BCP - ticket component: XX-S4C-OPR-SRV.

4 Download SAP S/4HANA Cloud Default Certificate

📘 Note

This procedure also applies to customer-managed certificate.

Procedure

1. Log on to SAP S/4HANA Cloud system with a user with the **Administrator** role.
2. Open the app [Maintain Client Certificate](#).
3. In [Certificate](#) field, input **Client Default** and search for this certificate. Click the download button in the [Actions](#) column.
4. In the popup screen [Export Certificate](#), select the default certificate format [Base-64-encoded X.509 certificate \(.crt\)](#) and click [Export](#) to download the .crt file.
The content of your downloaded certificate file is presented in a single line.

5 Set up Configuration Using Booster in SAP BTP Cockpit

Prerequisites

Before you start the booster on the SAP BTP cockpit, it's important that all prerequisites are met. These requisites notably include necessary permissions, app entitlements, availability of certain systems like SAP S/4HANA Cloud Public Edition, and an Identity Authentication Service (IAS) instance.

1. You are a global account Administrator.
2. You should have the entitlement to SAP Ariba Invoicing with one of the following plans in your SAP BTP Cockpit. For more information, see [Managing Entitlements and Quotas Using the Cockpit](#).
 1. test
 2. standard
 3. tandd
3. You should have the entitlement to SAP Central Invoice Management, Inbound Processing Module with standard (Application) plan.
4. You should have at least one Identity Authentication Service instance configured as part of your SAP BTP Cockpit global account.

Note

To verify the existence of Identity Authentication Service instances, log on to your SAP BTP global account and go to ► [Security](#) ► [Trust Configuration](#) ► [Establish Trust](#) ►.

You will find at least one tenant or a list of tenants in the new window if you have available instances.

Your user must have admin access to Identity Authentication Service before running the booster.

5. You should have at least one SAP S/4HANA Cloud Public Edition system registered in your SAP BTP cockpit under ► [System Landscape](#) ► [Systems](#) ►. For more information, see [Register an SAP S/4HANA Cloud Public Edition System in a Global Account](#).

When registering the SAP S/4HANA Cloud Public Edition system, for [Communication Scenario Groups](#), you must choose both the communication scenario groups [Integration with SAP Ariba Invoicing](#) and [Integration with SAP Master Data Integration](#).

Caution

Register the SAP S/4HANA Cloud Public Edition system even if there is an SAP S/4HANA Cloud Public Edition system listed under ► [System Landscape](#) ► [Systems](#) ► already.

If an SAP S/4HANA Cloud Public Edition system has already been used in a previous booster run, and is included in existing formations under ► [System Landscape](#) ► [Formations](#) ► in your SAP BTP account, do not reuse this SAP S/4HANA Cloud Public Edition system until the existing formations are deleted.

Procedure

1. Navigate to your global account in SAP BTP cockpit.
2. Choose [Booster Link](#) to complete the automatic setup steps in SAP BTP and SAP S/4HANA Cloud Public Edition.
3. The [Get Started with Set Up SAP Ariba Invoicing - Quick Account Setup](#) window displays.
4. Under section [Specify System Landscape](#), select appropriate values for the following fields:
 - [Select System Landscape Type](#): **Select your system landscape for SAP Ariba Invoicing subscription.**
 - [SAP S/4HANA Cloud Public Edition System](#): **Select the SAP S/4HANA Cloud Public Edition system you want to integrate SAP Ariba Invoicing with.**
 - [Identity Authentication System](#): **Select your tenant for Identity Authentication Service.**
5. Choose [Set Up My Account](#).

Note

Depending on connection conditions, the booster **Set Up SAP Ariba Invoicing** needs 30 minutes or longer to complete the account setup. Do not refresh your web page during the execution process, and make sure that your system is not locked or disconnected from the Internet while the booster is running, as this may result in failure.

If you encounter errors in step [Setting up subaccount\(s\)](#) during booster setup, click [Rerun](#). Booster execution will be triggered again and it can help to resolve any intermittent failures in the subaccount setup process. If the issue still persists after 3 retries, report an incident to the component [S2P-CIM-SET](#).

The [Rerun](#) option is not applicable to the failure in step [Setting up integrations](#). It is also not applicable when the generic error message **Setup encountered issues and could not be completed.** pops up.

To clean up previous configurations, see [Booster Cleanup Guide](#).

To solve issues, see [Booster Troubleshooting Guide](#).

If you need further assistance, raise an incident to the component [S2P-CIM-SET](#).

Results

After the successful execution of [Set Up SAP Ariba Invoicing](#) booster, new subaccounts, subscriptions, instances and destinations will be created in your SAP BTP Cockpit.

Two new subaccounts will be created in your SAP BTP Cockpit. One subaccount is dedicated to SAP Ariba Invoicing and another is dedicated to SAP Master Data Integration. More details can be found in the Booster success screen.

Table 2:

Subaccount	SAP Ariba Invoicing (CIM)	Master Data Integration (MDI)
Instances	SCIM Instance, S/4 Instance	
Destinations	SAP S/4HANA Cloud Public Edition, MDCS	S/4 HANA_BP, S/4 HANA_MDO

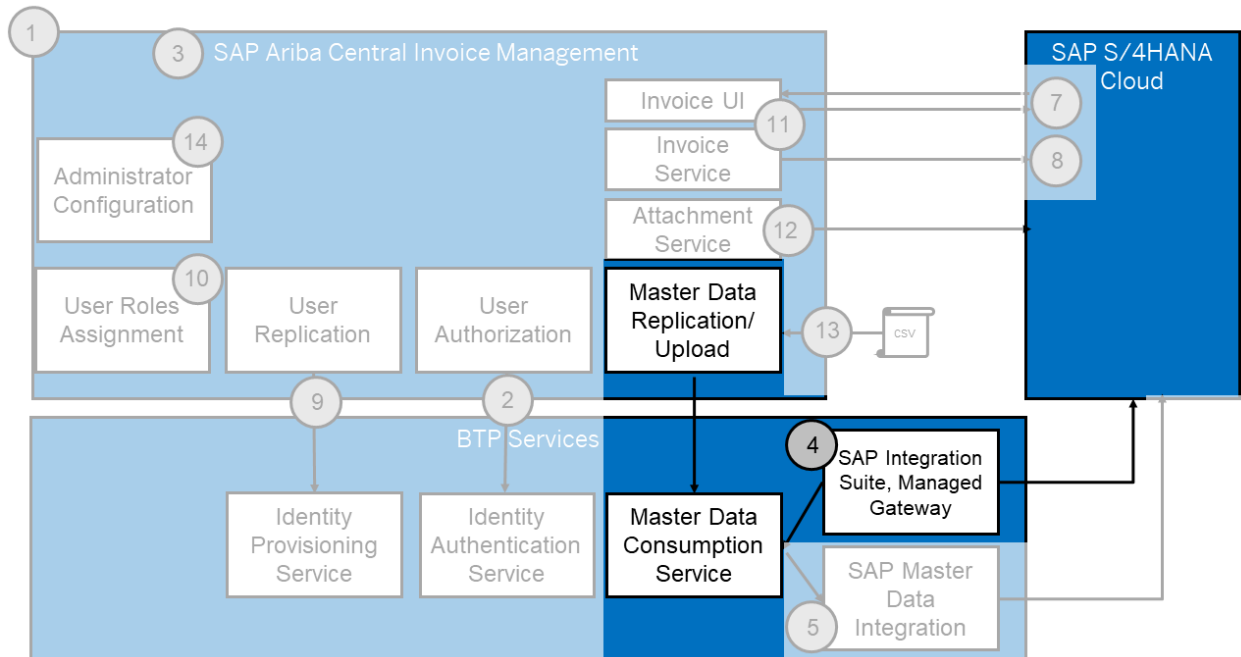
Subaccount	SAP Ariba Invoicing (CIM)	Master Data Integration (MDI)
Trust Setup	With IAS Host	
Environment	With Space	With Space
Entitlements	CIM, CIM-Inbound, CIM-SCIM	MDI
Subscriptions	CIM, CIM Inbound	MDO, MDI Tenant
Communication Arrangements in SAP S/4HANA Cloud Public Edition System	SAP_COM_0516	SAP_COM_0594
	SAP_COM_0897	SAP_COM_0659
		SAP_COM_0008
		SAP_COM_0091
Role Collections	Standard Role Collections for CIM	Business Configuration Admin, Master Data Orchestrator Admin
User Groups	Standard User Groups for CIM	

📘 Note

Information about user role assignment can be found at [Assigning Administrator Roles to Yourself or to the Administrator](#).

6 Master Data Integration via SAP Integration Suite, managed gateway

The steps involved in the entire configuration process are:



The following master data entities can be replicated from SAP S/4HANA Cloud Public Edition to SAP Ariba Invoicing:

- Company code
- G/L account
- Tax code

6.1 Setup SAP Integration Suite, managed gateway

Depending on your configuration environment, choose one of the following options:

Procedure in CIAS

This process to enable SAP Ariba Invoicing integration with SAP Integration Suite, managed gateway integration for master data is automated with an SAP Integration Suite, managed gateway API.

Please provide your customer name, BTP subaccount tenant ID, Email address and MDCS data center based on your contract.

📌 Note

The requested email address is used for receiving an email to activate your **SAP Integration Suite, managed gateway user ID (P-User ID)** for SAP Ariba Invoicing master data integration and to define its password. Each tenant must use a different P-User ID.

After the SAP Integration Suite, managed gateway user ID (P-User ID) has been created by SAP, in the inbox of the email address you input in CIAS procedure, you will receive an email from SAP IAS No Reply (ias@notifications.sap.com) with the title *Activate Your Account for User Profile*. The email content starts with "Dear <Customer Name> initial". You must click on the link included to activate your SAP Integration Suite, managed gateway user and set a password.

→ Tip

After activating your account and setting your password, you can find P-User ID in your user profile, under section *Personal Information* → field *User ID*.

In case you do not receive the activation email, or need to change the password, you can go to <https://aribaoperations.accounts.ondemand.com/> and use *Forgot password?* link with the email address you included in the incident.

Manual Procedure

📌 Note

We strongly recommend to create a P-user via automation. Please enter your value in the *Customer Name*, *BTP Subaccount ZoneID/TenantID* and *Email* fields in *Configuration Parameters* section, and click *Execute Task*.

Only follow the manual instructions in this chapter if you encounter issues after clicking *Execute Task*, for example, the email for P-user information is not sent to you after a few hours.

Please raise an SAP Support Portal incident ticket on component BNS-ARI-CI-PLT with a request to enable SAP Ariba Invoicing integration with SAP Integration Suite, managed gateway integration for master data. The subject of your ticket should be: **SAP Integration Suite, managed gateway user ID (P-User ID) for Central Invoice Management master data.**

The requested email address is used for receiving an SAP Integration Suite, managed gateway user ID (P-User ID) by email as well as the link to define its password. For each of your SAP Ariba Invoicing tenant (production tenant and test tenant), share your zone ID, data center, system ID and a valid email address. The P-user for each SAP Ariba Invoicing subaccount is unique. Please use a different email address for each P-user. Different P-users can't share a same email address.

You need SAP Integration Suite, managed gateway for master data to set-up inbound processing module of SAP Ariba Invoicing. The use of this software is included in your inbound processing module of SAP Ariba Invoicing. This inbound processing module is independent of any other SAP Ariba solutions. Therefore this ticket is mandatory even if you already have an account in SAP Integration Suite, managed gateway. In your ticket, please explicitly mention the following message, so that the context of the request is known to the processing team:

Enable SAP Ariba Invoicing integration with SAP Integration Suite, managed gateway for master data.

📌 Note

The requested email address is used for receiving an email to activate your **SAP Integration Suite, managed gateway user ID (P-User ID)** for SAP Ariba Invoicing master data integration and to define its password. Each tenant must use a different P-User ID.

Please follow the steps below on how to get the zone ID and system ID:

Zone ID:

- In SAP BTP Cockpit, click on your subaccount dedicated to SAP Ariba Invoicing.
- In the [Subaccount Overview](#) page, copy the tenant ID from the section [Subaccount Details](#).

System ID

- Log in to SAP S/4HANA Cloud Public Edition system and open app [Communication Systems](#).
- Copy the system ID from the row flagged as **Own System = Yes**

📌 Note

Production and test tenants will have different zone IDs and system IDs.

After the SAP Integration Suite, managed gateway user ID (P-User ID) has been created by SAP, in the inbox of the email address you included in the ticket, you will receive an email from SAP IAS No Reply (ias@notifications.sap.com) with the title *Activate Your Account for SAP Cloud Identity*. The email content starts with "Dear <System ID> initial". You must click on the link included to activate your SAP Integration Suite, managed gateway user ID and set a password.

→ Tip

After activating your account and setting your password, you can find P-user ID in your user profile, under section [Personal Information](#) → field [User ID](#).

In case you do not receive the activation email, or need to change the password, you can go to <https://aribaoperations.accounts.ondemand.com/> and use [Forgot password?](#) link with the email address you included in the ticket.

6.2 Configure the Connected System - SAP S/4HANA Cloud Public Edition

6.2.1 Create Communication User

Purpose

This procedure describes how to create an inbound communication user. You can either create a new user or use an existing one. If you're using an existing one, you can skip the following procedures.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an Administrator.
2. Open [Maintain Communication Users](#)-->[Maintain Communication Users](#) (F1338) app.
3. To create a new communication user, choose [New](#).
The [Create Communication User](#) screen is displayed.
4. On the [Create Communication User](#) screen, make the following entry:
Enter a user name, for example, **4N6_INBOUND**, a description, and a password in the respective fields.
5. Choose [Create](#).
The [Maintain Communication User](#) screen is displayed.

6.2.2 Create Communication System

Purpose

This procedure describes how to create the communication system that represents SAP Integration Suite, managed gateway system.

Procedure

1. Log on to the SAP Fiori launchpad as an **Administrator**.
2. Open the [Communication System](#) (F1762) app.

- To create a new communication system, choose [New](#).
The [New Communication System](#) dialog box is displayed.
- In the [New Communication System](#) dialog box, enter the following data according to your business requirements:

Field Name **Entry Value**

System ID **<Cloud_Integration_Gateway>**

System Name **<SN_Cloud_Integration_Gateway>**

Choose [Create](#).

- On the [Communication System](#) screen, go to the [General](#) tab. Enter the following data:

Field Name **Host Name** **Business System**

Value for Europe Data Center Enter the value **acig.ariba.com**. Enter the value **zone_uuid:<zone uuid>**.

Value for US Data Center Enter the value **acig-us.ariba.com** Enter the value **zone_uuid:<zone uuid>**.

Description Enter the host name of SAP Integration Suite, managed gateway. Please follow these steps to retrieve the parameter **<zone uuid>**:

- In SAP BTP Cockpit, click on your subaccount dedicated to SAP Ariba Invoicing.
- In the [Subaccount Overview](#) page, copy tenant ID from [Subaccount Details](#) section.

Business System Example:

zone_uuid:2efc05ce-6323-4449-806d-04bb29a772a0

ⓘ Note

The **zone_uuid** before the tenant ID is a must value. Don't remove it.

- On the [User for Inbound Communication](#) tab, choose [Add](#).
The [New Inbound Communication User](#) dialog box is displayed.
- In the [New Inbound Communication User](#) dialog box, enter the following data:
 - User Name:** Enter the communication user you created in [Create Communication User \[page 17\]](#), for example, **4N6_INBOUND**.

→ Tip

You can use the [Value Help](#) to search for a specific user name.

- Authentication Method:** From the dropdown list, choose [User Name and Password](#).

- Choose *OK*.
8. On the *User for Outbound Communication* tab, choose *Add*.
The *New Outbound User* dialog box is displayed.
 9. In the *New Outbound User* dialog box, enter the following data:
 - *Authentication Method*: From the dropdown list, choose *User Name and Password*.
 - *User Name*: **<SAP Integration Suite, managed gateway user ID (P user ID)>**
The P-user ID is case-sensitive.
 - *Password*: **<Password>**
- Choose *Create*.
10. Choose *Save*.

For master data integration, set up communication arrangements as described in the following subchapters.

6.2.3 Set Up Accounting Master Data Integration (SAP_COM_0179)

Use

This procedure describes how to set up the communication scenario SAP_COM_0179 for accounting master data integration via SAP Integration Suite, managed gateway. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to your SAP S/4HANA Cloud Public Edition system as an Administrator.
2. Open the *Communication Arrangements* (F1763) app.

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

3. To create a new communication arrangement, choose *New*.

4. In the *New Communication Arrangement* dialog box, enter the following data:

Field Name	Entry Value
<i>Scenario</i>	SAP_COM_0179 (Accounting Master Data Integration) → Tip You can use the <i>Value Help</i> to search for a specific scenario.
<i>Arrangement Name</i>	<Arrangement Name>

5. Choose *Create*.
6. On the *Communication Arrangements* screen, in the *Common Data* section, enter the following data according to your business requirements:

🔗 Example

Field Name	Entry Value	Description
<i>Communication System</i>	<Cloud_Integration_Gateway>	Select the communication system you created in Create Communication System [page 17] . After you selected the communication system, the value is also updated for the inbound communication, the inbound services, and the outbound communication.

- On the [Communication Arrangements](#) screen, in the [Outbound Services](#) section, enter the following data for the respective outbound service according to your business requirements:

Example

Outbound Services	Path	Replication Model Example Values
<i>Company Code - Send</i>	/cxf/receiveERPMD	<Ccode>  Note Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/". Enter the name for your own replication model.  Note Please maintain your value in <i>Package Size</i> field under both <i>Job Execution Details</i> section and <i>Additional Properties</i> section. This value must be higher than the total number of records that need to be uploaded. For example, if you have 500 company codes to be uploaded, set the value of <i>Package Size</i> to 500 or above.
<i>Replicate G/L Accounts</i>	/cxf/receiveERPMD	<GLAcc>  Note Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/". Enter the name for your own replication model.  Note Please maintain your value in <i>Package Size</i> field under <i>Job Execution Details</i> section. We suggest you to maintain the package size as 5,000.

Outbound Services	Path	Replication Model Example Values
<i>Tax Code Replication</i>	/cxf/receiveERPMD	<TaxCd> Note Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/". Enter the name for your own replication model. Note Please maintain your value in <i>Package Size</i> field under both <i>Job Execution Details</i> section and <i>Additional Properties</i> section. This value must be higher than the total number of records that need to be uploaded. For example, if you have 5,000 tax codes to be uploaded, set the value of <i>Package Size</i> to 5,000 or above.

- Make sure you only activate G/L account, company code and tax code in outbound services. Other items should be left unchecked. Enter dummy replication model values for other entities like profit center and cost center.
- We strongly recommend schedule regular jobs for *Company Code - Send*, *G/L Account Replication*, and *Tax Code Replication* once or twice a day. Running jobs more frequently than this may cause data flooding.

→ Tip

- If no regular job is scheduled, no master data will be sent to SAP Ariba Invoicing. Therefore, we require at least one initial replication by replication model.
- No master data will be received in SAP Ariba Invoicing before the first job is executed. If you need data early for project activities, you should run an initial replication by the replication model.

8. Choose *Save*.

The communication arrangement becomes active, and all required configurations are generated.

7 Add SAP Ariba Invoicing to your system HTTP Allowlist

To add SAP Ariba Invoicing in the HTTP allowlist of your system, please follow the instructions described in [Front-End Security](#).

8 User Replication

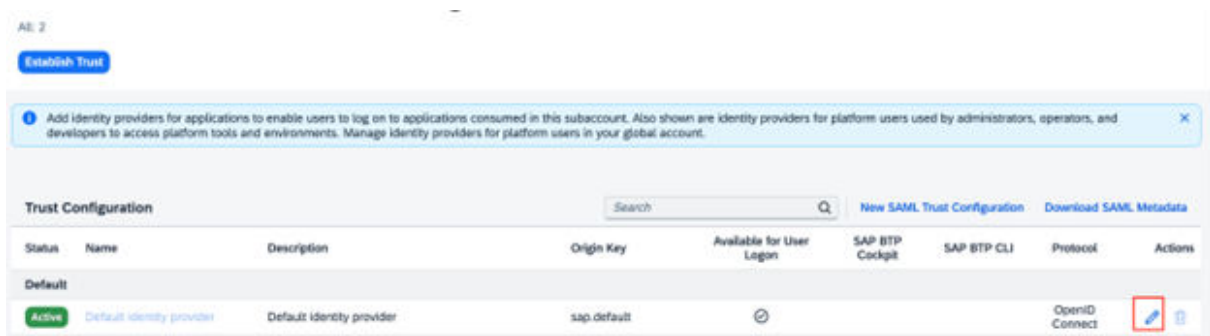
1.
 - **If you use SAP Cloud Identity Services, follow the steps below:**
Sign in to the administration console of SAP Cloud Identity Services and navigate to ► [Identity Provisioning](#) ► [Target Systems](#) ►.
 - **If you use Identity Provisioning Service in Neo Environment, access the Identity Provisioning UI by:**
 - [Access Identity Provisioning UI of Bundle Tenants](#)
 - [Access Identity Provisioning UI of Standalone Tenants](#)
2. Select the target system you created via booster. You can use CIM tenant ID to search it.
3. In the [Details](#), add a source system to read users from. The two most frequently used source systems are [Identity Authentication](#) and [SAP S/4HANA Cloud Public Edition](#). You can follow the steps below to create your source systems if there is none available.
 - [Source system for Identity Authentication](#)
 - [Source system for SAP S/4HANA Cloud Public Edition](#)
 - For other source systems, see [Source System](#).
4. **⚠ Caution**
Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you used in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see [Manage Job Notifications](#).

Start an identity provisioning job. Complete the steps in [Monitor Provisioning Job Logs](#).

9 Deactivate Default Identity Provider User Logon

Procedure

1. Log on to your SAP Ariba Invoicing subaccount in SAP BTP Cockpit. Choose ► [Security](#) ► [Trust Configuration](#) ►. In the [Trust Configuration](#) table, locate the row with name [Default Identity provider](#) and choose [Edit](#) in the [Action](#) column.



2. In the popup window [Edit Trust Configuration](#), uncheck the box [Available for User Logon](#). Choose [Save](#).

Edit Trust Configuration

Name:

Description:

Available for User Logon: ☐

Create Shadow Users During Logon: ☐

[Save](#) [Cancel](#)

3. Log on to the Identity Authentication service.
4. From the navigation area, choose [Applications & Resources Applications](#).
5. Search for the application that has been created as part of the trust setup.
The name of the application has the format [XSUAA_<Tenant ID>](#), but you can change it if needed.
6. Select your application.
Under [Single Sign-on](#) section, choose [Subject Name Identifier](#).
In [Subject Name Identifier](#) screen, under [Primary Attribute](#) section, set the [Value](#) field to [E-Mail](#) if this field is not maintained by default.

10 Create an External Tax Relevant Destination inside an SAP Ariba Invoicing Subaccount

Procedure

1. Get the following client credentials from third party vendors whom you are connecting for external tax calculation during external tax engine configuration.
 - Client ID
 - Client secret
 - URL
 - Token service URL
2. Log on to your SAP Ariba Invoicing subaccount created in [Create an SAP Subaccount in BTP for SAP Ariba Invoicing \[page 29\]](#), which can be completed by executing booster.
3. Click ► [Connectivity](#) ► [Destinations](#) ► in the left panel.
4. Click [New Destination](#) and fill the destination with following entries:

Field	Description
Name	Enter a name for this destination.
Type	Select HTTP .
Description	Optional. Enter a description for this destination.
URL	Enter the URL collected above.
Proxy Type	Select Internet .
Authentication	Select OAuth2ClientCredentials .
Client ID	Enter the client ID collected above.
Client Secret	Enter the client secret collected above.
Token Service URL Type	Choose the default value.
Token Service URL	Enter token service URL collected above.

5. In *Additional Properties* section, click *New Property*. Enter **sap.external.tax** in the field from the dropdown list, and make the following entries:

Property	Value	Mandatory	Description
sap.external.tax	Name of the external tax engine maintained in the source column during configuration. See API for Tax Engine Configuration Based on Company Code for more information.	Yes	External tax consumption destination

6. Click *Save*.

11 Further Operations

Once you have configured SAP Ariba Invoicing, you or the administrator have to follow instructions described in [SAP Ariba Invoicing Administration Guide Introduction](#).

12 Appendix

⚠ Caution

This chapter includes all the configuration steps that are covered by the booster **Set Up SAP Ariba Invoicing**. You can use this section as a reference for manual configuration.

Only check this step when booster **Set Up SAP Ariba Invoicing** fails.

12.1 SAP Ariba Invoicing Solution Description

With the [SAP Ariba Invoicing](#), you can have three SAP Ariba Invoicing tenants: one tenant for production, and up to two tenants for testing. You need to create one different SAP BTP subaccount for each of these tenants.

With the [Test, Demo and Development](#) license, you can have one SAP Ariba Invoicing tenant. You can use this tenant for test, demo and development purpose. You must create one SAP BTP subaccount for this tenant.

12.2 SAP BTP Cockpit Configuration Procedure

The following section describes configuration steps you need to complete in SAP BTP Cockpit. These steps are the prerequisites of this setup.

12.2.1 Create an SAP Subaccount in BTP for SAP Ariba Invoicing

Prerequisites

→ Recommendation

With the [SAP Ariba Invoicing](#) license, you are allowed to create three SAP BTP subaccounts: one for your production tenant, and up to two for your test tenants.

With the *Test, Demo and Development* license, you only need to create one SAP BTP subaccount for your test, demo and development tenant.

Read [Setting Up Your Account Model](#) and [Account Administration in the Cockpit](#) for information on subaccount management.

Create a Subaccount

Prerequisites

You are a global account administrator.

Context

You create subaccounts in your global account. Once you create a new subaccount, you see a tile for it in the global account view, and you are automatically assigned to it as an administrator.

Procedure

1. From your global account, navigate to the [Account Explorer](#) page.
2. Select ► [Create](#) ► [Subaccount](#) ►.
3. Specify a display name.
4. **Optional:** Enter a description.
5. Enter a subdomain for your subaccount. This will become part of the URL for accessing applications that you subscribe to from this subaccount.

Note

The subdomain can contain up to 63 characters such as letters, digits and hyphens (not allowed in the beginning or at the end), and must be unique across all subaccounts in the same region. Uppercase and lowercase letters can be used, however that alone does not qualify as a means to differentiate subdomains (e.g. **SUBDOMAIN** and **subdomain** are considered to be the same).

6. Choose the desired region for your subaccount.
7. Select a parent for your subaccount. The parent can either be the global account or a directory.
8. If your subaccount is to be used for production purposes, select the [Used for production](#) option.

Note

This does not change the configuration of your subaccount. Use this flag for your internal use to operate your production subaccounts in your global account and systems more efficiently. Your cloud operator may

also use this flag to take appropriate action when handling incidents related to mission-critical accounts in production systems.

You can change your selection at any time by editing the subaccount properties. Do not select this option if your account is used for non-production purposes, such as development, testing, and demos.

9. **Optional:** To use beta services and applications in the subaccount, under [Advanced](#), select [Enable beta features](#).

⚠ Caution

You shouldn't use SAP BTP beta features in subaccounts that belong to productive enterprise accounts. For more information, see [Important Disclaimers and Legal Information](#).

📌 Note

Once you have enabled this setting in a subaccount you cannot disable it.

10. **Optional:** Under ► [Advanced](#) ► [Labels](#) ►, assign labels to the subaccount to make organizing and filtering your subaccounts easier. For more information, see [Labels](#).

→ Tip

- When adding multiple values to a label, press after each value.
- When you start typing, any matching label names and values that are already assigned to other subaccounts in your global account are offered as suggestions.

11. Save your changes.

Results

A new tile appears in the global account page with the subaccount details.

Next Steps

Once you've created your subaccount, navigate to it to enable the environment that you wish to use.

12.2.2 Create an Environment Instance or Organization

Enable an environment or create an environment instance using the SAP BTP cockpit.

Prerequisites

- You're managing a multi-environment subaccount.
- You're a subaccount administrator.
- You have free quota assigned to your subaccount.

Context

On a subaccount level, environments constitute the actual platform-as-a-service (PaaS) offering of SAP BTP that allows for the development and administration of business applications.

Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. The availability of different environments allows for greater flexibility in your development process.

For more information, see [Enviroments](#).

Procedure

1. Navigate into the subaccount.
2. Choose your method: *Enable Environment* or *Create an Environment Instance*.
 - *Enable Environment*: On the *Subaccount - Overview* page, click *Enable <Environment>*.
 - *Create an Environment Instance*: See [Environment Instances](#).

Results

You can use the environment or environment instance in your subaccount.

12.2.3 Create a BTP Space

Create spaces in your Cloud Foundry organization using the SAP BTP cockpit.

Prerequisites

You have a Cloud Foundry organization in your Cloud Foundry subaccount, and you have the Org Manager role in that organization.

See [Create Orgs](#).

Procedure

1. Navigate to the subaccount that contains the Cloud Foundry organization in which you'd like to create a space.
2. On the subaccount [Overview](#) page, you have a [Spaces](#) table. Choose [Create Space](#) from the top right hand-side corner of that table.
3. Enter a space name and choose the permissions you'd like to assign to your ID, then choose [Save](#).

Next Steps

To assign quota to spaces, see [Change Space Quota Plans](#).

You can change the name of a space by going to the [Cloud Foundry > Spaces](#) page from the left hand-side navigation and choosing  (Edit) on the tile of that space. You can also create additional spaces from there.

Related Information

[Create Orgs](#)

[Create Spaces Using the Cloud Foundry Command Line Interface](#)

12.2.4 Establish an OpenID Connect Trust

You must set up a trust between your subaccount and the Identity Authentication service. It will allow you to subscribe to SAP Ariba Invoicing and provide authentication and single sign-on service for your users to your SAP Ariba Invoicing tenant. If needed, Identity Authentication service can be configured to delegate authentication to a third party identity provider.

Procedure

1. In the SAP BTP cockpit, go to the subaccount for your SAP Ariba Invoicing system.
2. From the navigation area, choose [Security Trust Configuration](#).
3. Choose [Establish Trust](#).
4. In the following popup window, select your test or productive Identity Authentication tenant from the dropdown list. Choose [Next](#).
Only Identity Authentication tenants that are associated with your customer ID will be displayed.
5. In the step [Choose Domain](#), select a domain from the dropdown list. Choose [Next](#).
If your SAP S/4HANA Cloud Public Edition tenant URL is formatted [https://<mytenant>.s4hana.cloud.sap](#), select the domain ends with [accounts.cloud.sap](#), or the domain ends with [accounts.ondemand.com](#).

6. In step **Configure Parameter**, you can update the parameters in all fields except for **Origin Key**, or keep the default values. Then choose **Next**.
7. In step [Review](#), review your settings. Choose [Previous](#) if you want to change previous settings. Choose [Finish](#) to complete your configuration.

Note

The [Default Identity Provider](#) in trust configuration should be deactivated for user login. As Default Identity Provider is a part of SAP ID services, it is not under customer control. If it is not deactivated, users will be asked to choose between different identity providers.

12.2.5 Entitlements to SAP Ariba Invoicing in SAP BTP Cockpit

Procedure

1. Navigate into the subaccount where you would like to configure the entitlements.
2. Choose [Entitlements](#) from the left navigation panel to see the entitlements for your subaccount.
3. Choose [Configure Entitlements](#).
4. Choose [Add Service Plans](#).

- For service, select [SAP Ariba Invoicing](#).
- For plan:

📘 Note

- With SAP Ariba Invoicing, you can create two test tenants and one production tenant. When subscribing to the SAP Ariba Invoicing service in the configuration procedure below, you must use the [test](#) plan for the test tenants and the [standard](#) plan for the production tenant.
- With the [Test, Demo and Development](#) license, you can create one test, demo and development tenant. When subscribing to the SAP Ariba Invoicing service in the configuration procedure below, you must use the [tandd](#) plan.

Select [test](#) to create a test tenant for non-productive purposes. This allows you to pre-validate configuration and test the application with a small number of invoices without impact to your subscription/license metric.

The test tenant also allows you to test upcoming features before they are activated in production. To show or hide the documentation related to the upcoming features, you need to use the [Feature release](#) filter above the page content on SAP Help Portal.

In addition, you can also use the [test](#) plan to create a second test tenant that you can connect to your project line systems.

⚠ Caution

Productive usage of the test tenants (with [test](#) plan assigned) is not allowed.

- Select [standard](#) to create a productive tenant.

⚠ Caution

The productive tenant (with standard plan assigned) is included in your subscription/license metric.

- Select [tandd](#) to create a test, demo and development tenant.

⚠ Caution

The test, demo and development tenant (with [tandd](#) plan assigned) is included in the subscription/license metric of your [Test, Demo and Development](#) license.

5. Once you're happy with the changes, choose [Add 1 Service Plans](#), then choose [Save](#).

12.2.6 Subscribe to SAP Ariba Invoicing in SAP BTP Cockpit

Procedure

1. In the [Service Marketplace](#), select the [SAP Ariba Invoicing](#) tile and click [Create](#).
2. In the [New Instance](#) or [Subscription](#) window,
 - For service, select [SAP Ariba Invoicing](#).

- For plan:

Note

- With the SAP Ariba Invoicing, you can create one test tenant and one production tenant. When subscribing to the SAP Ariba Invoicing service in the configuration procedure below, you must use the [test](#) plan for the test tenant and the [standard](#) plan for the production tenant.
- With the [Test, Demo and Development](#) license, you can create one test, demo and development tenant. When subscribing to the SAP Ariba Invoicing service in the configuration procedure below, you must use the [tandd](#) plan.

- If you want to create a test tenant for non-productive purposes, select [test](#). This allows you to pre-validate configuration and test the application with a small number of invoices without impact to your subscription/license metric.

Caution

Productive usage of the test tenant (with test plan assigned) is not allowed.

3. If you want to create a productive tenant, select [standard](#).

Caution

The productive tenant (with standard plan assigned) is included in your subscription/license metric.

Once the subscription is complete, an application named SAP Ariba Invoicing is automatically created in the IAS tenant. Please do not remove this application while your subscription is active.

4. If you want to create a test, demo and development tenant, select [tandd](#).

Caution

The test, demo and development tenant (with tandd plan assigned) is included in the subscription/license metric of your test, demo and development license.

12.3 SAP S/4HANA Cloud Public Edition Administration and Configuration Roles

The following roles are provided as example roles from SAP. You can use these as templates to create your own roles.

Business Roles

Business Role Name	Business Role ID As Delivered by SAP
Administrator	SAP_BR_ADMINISTRATOR
BPC Expert	SAP_BR_BPC_EXPERT

12.4 Master Data Integration

Master data has to be replicated from the connected systems to SAP Ariba Invoicing on SAP Business Technology Platform. The following master data entities have to be replicated:

1. Business partner (supplier)
2. Project controlling object (WBS element)
3. Cost center

The master data entities will be replicated from SAP S/4HANA Cloud Public Edition through Master Data Integration service on SAP BTP to SAP Ariba Central Invoice Management on SAP BTP.

12.4.1 Master Data Integration via SAP Master Data Integration Service

→ Tip

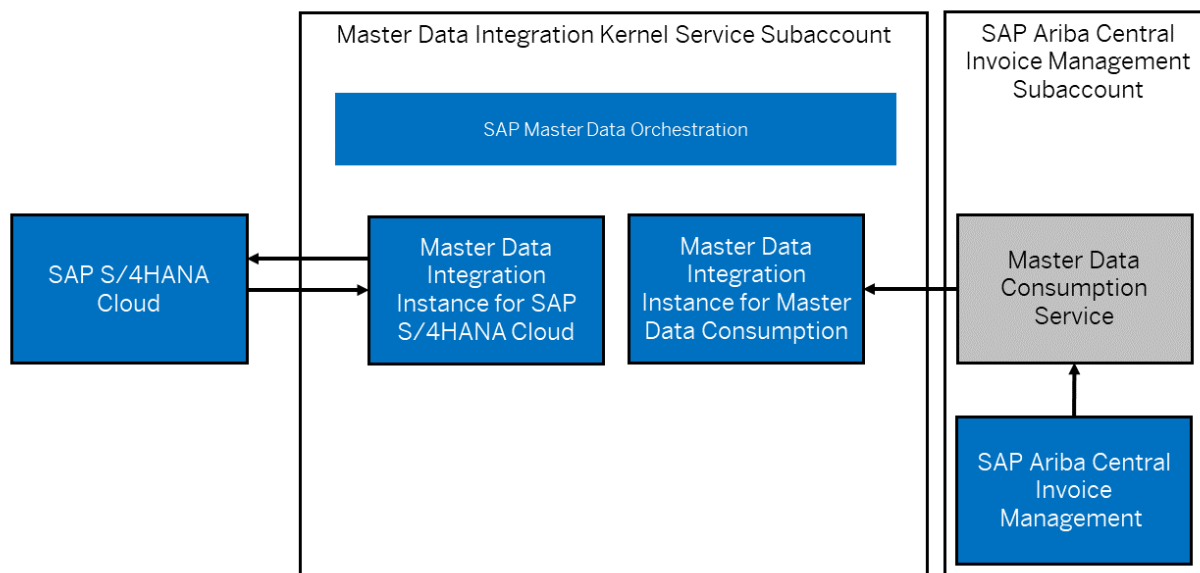
We have prepared two 2-minute videos for an overall introduction of SAP Master Data Integration Service. See:

[Introducing SAP Master Data Integration Service and Its Guiding Principles](#)

[Introducing Business Data Orchestration and Distribution Models](#)

The entire configuration process described in the diagram below.

SAP Master Data Integration service is a multi-tenant cloud service for master data integration. It provides a consistent view on master data across a hybrid landscape. Please also refer to [What Is Master Data Integration?](#) for more information.



SAP Business Data Orchestration is a service that runs on the SAP Business Technology Platform. This service is used as a central place to configure the distribution of master data. For more information, please refer to [General Setup of SAP Business Data Orchestration](#).

These services are used to replicate the following master data entities from SAP S/4HANA Cloud Public Edition into SAP Ariba Invoicing via Master Data Consumption service:

- Business Partner (Supplier)
- Project Controlling Object (WBS Element)
- Cost Center

You need to create:

- One SAP Master Data Integration kernel service subaccount that is in the same BTP global account as the one used for SAP Ariba Invoicing.
- One SAP Business Data Orchestration subscription in the master data integration kernel service subaccount

Note

You only need one kernel service subaccount, one instance for master data integration, and one SAP Business Data Orchestration subscription for each landscape in the SAP BTP global account (e.g. for test landscape or for production landscape). If they were already created for the integration of other applications in the targeted landscape, you don't need to create new ones and they can be reused.

For each SAP Ariba Invoicing tenant, you need to create one SAP Master Data Integration instance for read operation from SAP Master Data Integration to Master Data Consumption service in the same kernel subaccount. This instance is specific to the SAP Invoicing tenant. Once created, it can be reused if you need to connect more than one backend system to the same SAP Ariba Invoicing tenant.

For each SAP S/4HANA Cloud Public Edition system to be connected to the same SAP Ariba Invoicing, you need to:

- Create one specific SAP Master Data Integration instance for the SAP S/4HANA Cloud Public Edition system in the same kernel subaccount. If the instance was previously created to connect the SAP S/4HANA Cloud Public Edition system with another application, you can reuse it.
- Configure distribution models in SAP Business Data Orchestration.
 - One push and one pull replication models to read from SAP S/4HANA Cloud Public Edition to SAP Master Data Integration for each of these master data entities: project controlling object and cost center.
 - One pull distribution model to read from SAP Master Data Integration to Master Data Consumption service for each of these master data entities: business partner, project controlling object and cost center.

12.4.1.1 Configure the SAP Master Data Integration Service

The Master Data Integration service must be configured for the replication of Business Partner (Supplier) master data from SAP S/4HANA Cloud Public Edition to SAP Ariba Invoicing on SAP BTP.

→ Tip

You can refer to [Master Data Replication: Setup Instructions for SAP S/4HANA Cloud Public Edition](#) for the latest and comprehensive updates of SAP Master Data Integration configuration steps.

12.4.1.1.1 Create a Subaccount for SAP Master Data Integration Kernel Service

Create a Subaccount

Prerequisites

You are a global account administrator.

Context

You create subaccounts in your global account. Once you create a new subaccount, you see a tile for it in the global account view, and you are automatically assigned to it as an administrator.

Procedure

1. From your global account, navigate to the [Account Explorer](#) page.
2. Select ► [Create](#) ► [Subaccount](#) ►.
3. Specify a display name.
4. **Optional:** Enter a description.
5. Enter a subdomain for your subaccount. This will become part of the URL for accessing applications that you subscribe to from this subaccount.

📘 Note

The subdomain can contain up to 63 characters such as letters, digits and hyphens (not allowed in the beginning or at the end), and must be unique across all subaccounts in the same region. Uppercase and lowercase letters can be used, however that alone does not qualify as a means to differentiate subdomains (e.g. **SUBDOMAIN** and **subdomain** are considered to be the same).

6. Choose the desired region for your subaccount.
7. Select a parent for your subaccount. The parent can either be the global account or a directory.
8. If your subaccount is to be used for production purposes, select the [Used for production](#) option.

📘 Note

This does not change the configuration of your subaccount. Use this flag for your internal use to operate your production subaccounts in your global account and systems more efficiently. Your cloud operator may also use this flag to take appropriate action when handling incidents related to mission-critical accounts in production systems.

You can change your selection at any time by editing the subaccount properties. Do not select this option if your account is used for non-production purposes, such as development, testing, and demos.

9. **Optional:** To use beta services and applications in the subaccount, under [Advanced](#), select [Enable beta features](#).

⚠ Caution

You shouldn't use SAP BTP beta features in subaccounts that belong to productive enterprise accounts. For more information, see [Important Disclaimers and Legal Information](#).

📘 Note

Once you have enabled this setting in a subaccount you cannot disable it.

10. **Optional:** Under ► [Advanced](#) ► [Labels](#) ▾, assign labels to the subaccount to make organizing and filtering your subaccounts easier. For more information, see [Labels](#).

→ Tip

- When adding multiple values to a label, press `Enter` after each value.
- When you start typing, any matching label names and values that are already assigned to other subaccounts in your global account are offered as suggestions.

11. Save your changes.

Results

A new tile appears in the global account page with the subaccount details.

Next Steps

Once you've created your subaccount, navigate to it to enable the environment that you wish to use.

12.4.1.1.2 Create an Environment Instance or Organization

Enable an environment or create an environment instance using the SAP BTP cockpit.

Prerequisites

- You're managing a multi-environment subaccount.

- You're a subaccount administrator.
- You have free quota assigned to your subaccount.

Context

On a subaccount level, environments constitute the actual platform-as-a-service (PaaS) offering of SAP BTP that allows for the development and administration of business applications.

Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. The availability of different environments allows for greater flexibility in your development process.

For more information, see [Enviroments](#).

Procedure

1. Navigate into the subaccount.
2. Choose your method: [Enable Environment](#) or [Create an Environment Instance](#).
 - [Enable Environment](#): On the [Subaccount - Overview](#) page, click [Enable <Environment>](#).
 - [Create an Environment Instance](#): See [Environment Instances](#).

Results

You can use the environment or environment instance in your subaccount.

12.4.1.1.3 Create a BTP Space

Create spaces in your Cloud Foundry organization using the SAP BTP cockpit.

Prerequisites

You have a Cloud Foundry organization in your Cloud Foundry subaccount, and you have the Org Manager role in that organization.

See [Create Orgs](#).

Procedure

1. Navigate to the subaccount that contains the Cloud Foundry organization in which you'd like to create a space.
2. On the subaccount [Overview](#) page, you have a [Spaces](#) table. Choose [Create Space](#) from the top right hand-side corner of that table.
3. Enter a space name and choose the permissions you'd like to assign to your ID, then choose [Save](#).

Next Steps

To assign quota to spaces, see [Change Space Quota Plans](#).

You can change the name of a space by going to the [Cloud Foundry > Spaces](#) page from the left hand-side navigation and choosing  (Edit) on the tile of that space. You can also create additional spaces from there.

Related Information

[Create Orgs](#)

[Create Spaces Using the Cloud Foundry Command Line Interface](#)

12.4.1.1.4 Create SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition

12.4.1.1.4.1 Create Service Instance

Purpose

This procedure describes how to create a master data integration service instance.

Procedure

1. Log on to your SAP BTP subaccount dedicated to Master Data Integration kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).

3. To create a new instance of the master data integration service, choose [Create](#).
The [New Instance or Subscription](#) dialog box is displayed.
 - **Service:** From the dropdown list, choose [Master Data Integration](#).
 - **Plan:** From the dropdown list, choose [sap-integration](#).
 - **Runtime Environment:** From the dropdown list, choose [Cloud Foundry](#).
 - **Space:** Select any option from the dropdown list. The chosen space is not relevant for master data integration service.
 - **Instance Name:** Enter a service instance name, for example **<MDI_Service_Instance>**
4. Choose [Next](#).
5. On the next screen, enter the following parameters in JSON format:

Sample Code

```
{
  "businessSystemId": "<Business_System_Name>", "application":
  "s4", "writePermissions": [ { "entityType":
  "sap.odm.businesspartner.BusinessPartnerRelationship" }, { "entityType":
  "sap.odm.businesspartner.BusinessPartner" }, { "entityType":
  "sap.odm.businesspartner.ContactPersonRelationship" }, { "entityType":
  "sap.odm.finance.costobject.ProjectControllingObject" }, { "entityType":
  "sap.odm.finance.costobject.CostCenter" }, ] }
```

- **Business_System_Name:** Enter the logical system identifier of the client system that you will connect to the Master Data Integration instance. In SAP S/4HANA Cloud Public Edition, you can access app [Communication Systems](#), use the filter [Own SAP Cloud System](#) and find this value in communication system details.

Note

Master Data Integration only allows a system name no longer than 60 characters.

If you enter a name in Master Data Integration that is not the same as it is in SAP S/4HANA Cloud Public Edition, the process would not work properly.

6. Choose [Create](#). The new instance that is created is visible in [Instances](#).

12.4.1.1.4.2 Create and Download the Public Key of an SAP BTP Managed Certificate

Purpose

You can create a certificate at the destination level of SAP BTP cockpit. This certificate will be rotated by SAP Cloud Public Key infrastructure.

Procedure

1. Log on to your subaccount in SAP BTP Cockpit.
2. Go to ► [Connectivity](#) ► [Destinations](#) ► [Certificates](#) ►, and click [Generate Certificate](#) in the popup screen.
3. On [Generate new certificate](#) screen, make the following entries:
 - [Certificate File Name](#): **Your certificate file name**, for example, **cim**.
 - [File Name Extension](#): **P12**, **JKS** or **PFX**, depending on the context.
 - [Certificate Common Name](#) (optional): **Your certificate common name**.
 - [Certificate Validity Time Unit](#): **The time unit appropriate for your scenario**.
 - [Certificate Validity Value](#): **Appropriate validity value**.
 - [Password](#) (optional): **Your password**.
 - Check the box [Enable automatic renewal](#).
4. Click [Generate Certificate](#). This step will create an entry named <**Certificate File Name**>.p12, <**Certificate File Name**>.jks or <**Certificate File Name**>.pfx.
5. Click [Cancel](#).
6. Go to ► [Connectivity](#) ► [Destinations](#) ► [Certificates](#) ►, and click the newly generated certificate.
7. Select the certificate you created in step 4 and click [Export](#) for SAP S/4HANA Cloud upload.

12.4.1.1.4.3 Create Service Key

Prerequisites

⚠ Caution

This activity needs to be executed in an SAP BTP subaccount dedicated to MDI Kernel Service.

Purpose

Create a service key in the newly created service instance to obtain credentials to connect a client to the master data integration service.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service](#) [page 39].

2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Select the service instance you created (for example **<MDI_Service_Instance>**).
4. Under [Service Keys](#), choose [Create](#). A wizard opens, offering you to configure your new service key.
[Service Key Name](#): enter a service key name.

The service key contains credentials to authenticate a client in the master data integration service. Multiple service keys can be created, but will appear as one client in the master data integration service.

It is recommended to indicate the date of creation in the service key name, so that in case of security incidents it is easily possible to judge whether a specific set of credentials is compromised. For example: the recommended service key name for a service key created on Jan 1st 2020 would be *ValidFrom_20200101*.

5. Maintain the following value based on your authentication method:

Note

It is strongly recommended to use authentication method with mTLS certificate. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. **Basic Authentication Method:**

No additional parameter needed.

2. **mTLS Certificate Based Authentication Method:**

Follow the steps described in [Download SAP S/4HANA Cloud Default Certificate \[page 10\]](#).

Upload the following JSON payload:

Sample Code

```
{
  "xsuaa": {
    "credential-type": "x509",
    "x509": {
      "certificate": "[Content of downloaded the certificate file]-----BEGIN CERTIFICATE-----",
      "ensure-uniqueness": false,
      "certificate-pinning": false
    }
  }
}
```

6. Choose [Create](#).

Note

You can refer to [Connecting Clients](#) for additional details about service instance creation and service key creation.

12.4.1.1.4.4 Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition

Prerequisites

⚠ Caution

This activity needs to be executed in an SAP BTP subaccount dedicated to MDI Kernel Service.

Purpose

This procedure describes how to retrieve the service key. The service key contains the necessary information to authenticate at the APIs of the Master Data Integration service. This information needs to be copied into SAP S/4HANA Cloud Public Edition so that it can authenticate at the APIs of the Master Data Integration service. Steps on how to maintain this information in SAP S/4HANA Cloud Public Edition can be found in chapter [Configuration in SAP S/4HANA Cloud Public Edition \[page 59\]](#).

📘 Note

It is recommended to create a new service key and copy new credentials to SAP S/4HANA Cloud Public Edition in regular intervals. Once your new service key and credentials are in place, please delete the old credentials in SAP S/4HANA Cloud Public Edition, and delete the old service key.

You can refer to [Create Service Key \[page 44\]](#) for service key creation details.

There will probably be a downtime in SAP S/4HANA Cloud Public Edition system during the credential changing process.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Select the service instance you created (for example **<MDI_Service_Instance>**).
4. Under [Service Keys](#), choose [Actions](#) [View](#). Look up the relevant fields according to your authentication method.

📘 Note

It is strongly recommended to use authentication method with mTLS certificate. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. Basic Authentication Method

- *clientid*
- *clientsecret*
- *url*
- *identityzone*

Caution

The relevant field is **identityzone**, not *identityzoneid*. Even though *identityzoneid* is also in the payload and right after *identityzone*, it is not relevant.

- *uri*

2. mTLS Certificate Based Authentication Method

- *clientid*
- *certurl*
- *identityzone*
- *uri*

Please note them down as you will need to input these values later in SAP S/4HANA Cloud Public Edition and MDI configurations.

12.4.1.1.5 Create SAP Master Data Integration for Master Data Consumption Service

This activity needs to be executed in an SAP BTP subaccount dedicated to MDI Kernel Service.

12.4.1.1.5.1 Create Service Instance

Purpose

This procedure describes how to create a master data integration service instance for READ operation from SAP Master Data Integration to Master Data Consumption.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under *Services*, click on *Instances and Subscriptions*.

3. To create a new instance of the master data integration service, choose [Create](#). The [New Instance or Subscription](#) dialog box is displayed.
 - **Service**: From the dropdown list, choose [Master Data Integration](#).
 - **Plan**: From the dropdown list, choose [sap-integration](#).
 - **Runtime Environment**: From the dropdown list, choose [Cloud Foundry](#).
 - **Space**: Select any option from the dropdown list. The chosen space is not relevant for master data integration service.
 - **Instance Name**: Enter a service instance name, for example **<MDI_Service_Instance_MDCS>**
4. Choose [Next](#).
5. On the next screen, enter the following parameters in JSON format:

Sample Code

```
{
  "businessSystemId": "MDCS",
  "application": "ariba"
}
```

6. Choose [Create](#). The new instance that is created is visible in [Instances](#).

12.4.1.1.5.2 Create Service Key

Purpose

Create a service key in the newly created service instance to obtain credentials to connect a client to the master data integration service.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Select the service instance you created (for example **<MDI_Service_Instance_MDCS>**).
4. Under [Service Keys](#), choose [Create](#). A wizard opens, offering you to configure your new service key.
Service Key Name: Enter a service key name. The service key contains credentials to authenticate a client in the master data integration service. Multiple service keys can be created, but will appear as one client in the master data integration service. It is recommended to indicate the date of creation in the service key name, so that in case of security incidents it is easily possible to judge whether a specific set of credentials is compromised. For example: the recommended service key name for a service key created on Jan 1st 2020 would be ValidFrom_20200101.

5. Choose [Create](#).

12.4.1.1.5.3 Retrieve Client Credentials for Master Data Consumption

Purpose

This procedure describes how to retrieve the service key. The service key contains the necessary information to authenticate at the APIs of the Master Data Integration service. This information needs to be copied into the destination to Master Data Consumption inside SAP Ariba Invoicing subaccount so that it can authenticate at the APIs of the Master Data Integration service. Steps on how to maintain this information in that destination can be found in chapter [Create a Destination inside SAP Ariba Invoicing Subaccount \[page 58\]](#).

Note

It is recommended to create a new service key and copy new credentials to the destination to Master Data Consumption inside SAP Ariba Invoicing subaccount in regular intervals. Once your new service key and credentials are in place, please delete the old credentials in the destination to Master Data Consumption inside SAP Ariba Invoicing subaccount and delete the old service key.

You can refer to [Create Service Key \[page 48\]](#) for service key creation details.

There will probably be a downtime in Master Data Consumption system during the credential changing process.

Procedure

1. Log on to your SAP BTP subaccount dedicated for Master Data Integration kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Select the service instance you created (for example **<MDI_Service_Instance_MDCS>**).
4. Under [Service Keys](#), choose **► Actions ► View ►**.
5. Please note the following values down as you will need to input these values later in SAP S/4HANA Cloud Public Edition and MDI configurations.

Caution

The relevant field is **identityzone**, not **identityzoneid**. Even though **identityzoneid** is also in the payload and right after **identityzone**, it is not a relevant field.

- *clientid*
- *clientsecret*

- [url](#)
- [identityzone](#)
- [uri](#)

12.4.1.1.6 Configure Master Data Integration and Create Distribution Models

12.4.1.1.6.1 Subscribe to SAP Business Data Orchestration

This procedure describes how to subscribe to the SAP Business Data Orchestration application.

Procedure

Please carry out the steps to subscribe to the SAP Business Data Orchestration as described in [Subscription Process for SAP Business Data Orchestration](#).

12.4.1.1.6.2 Maintain Master Data Integration Configuration

Procedure

ⓘ Note

This configuration must be carried out with caution because the configuration options will no longer be available after it is activated.

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Select the service instance you created in [Create Service Instance \[page 42\]](#).
4. Under [Service Keys](#), choose [ActionsView](#). The relevant field is identityzone.
5. Add roles to a role collection following the steps described in [Add Roles to a Role Collection](#).
 1. Choose [Security Role Collections](#).
 2. To add roles, choose the role collection name.
 3. Go to the [Roles](#) section and choose [Edit](#).
 4. To add a role to the role collection, choose the input field. The role selection screen opens, just select the role BusinessConfigurationAdmin.

- If you accidentally deleted the input field for the new roles, choose [Add a role](#) to display a new input field.
5. Use the dropdown list or the [F4](#) function key under [Role Name](#) to display the roles that are available. The roles are sorted alphabetically. Choose the role `BusinessConfigurationAdmin`.
 6. Choose [Add](#).
 7. Save your changes.
6. Assign users to role collections using the steps defined in **Assign Users to Role Collections** section of Authentication in SAP Master Data Integration.
 1. Choose [Security Role Collections](#).
 2. Choose the role collection to which you want to assign users.
 3. Go to the [Users](#) section and choose [Edit](#).
 4. Enter the user ID of the user that you want to assign to the role collection. If the user only exists in a connected identity provider, you must choose the identity provider and type in the e-mail address.
 5. (Optional) To add more users, choose + (Add a user).
 6. Save your changes.

You've now assigned this user to a role collection. The user has all of the authorizations of this role collection.
 7. Go to [Instances and Subscriptions](#) in your subaccount. Under [Subscriptions](#), click [Master Data Integration \(Orchestration\)](#).
 8. Select [Go to Application](#). The system will navigate you to the fiori launchpad for SAP Business Data Orchestration.
 9. Go to section [Configure Master Data Integration](#) and select [Configure Business System Name](#) application.
 10. Click on [Create](#) button, the system will navigate you to the [Configure Business System Name](#) screen. Fill the [Business System](#) field with the identityzone value in the subaccount credential key, and click [Create](#).
 11. Go back to the home page and select [Configure SOAP Replication](#) application from the [Configure Master Data Integration](#) section. In the [Configure SOAP Replication](#) screen, check the box under [SOAP Business Partner Active](#). Click [Save](#).

12.4.1.1.6.3 Configure Distribution Model in SAP Business Data Orchestration for Business Partner

Purpose

This procedure describes how to maintain the distribution model for the Business Partner (Supplier).

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Choose [Master Data Integration \(Orchestration\)](#) from the [Subscriptions](#) section.

4. Choose [Go to Application](#).
A new window opens, asking you to select a login option.
5. Choose [Default Identity Provider](#).
The Business Data Orchestration home page is displayed.
6. Choose [Manage Distribution Model](#).
The [Manage Distribution Model](#) app is displayed.
7. Choose [Create](#).
8. On the next screen, enter the following data under [Header](#):
 - **Model: <BP - MDCS>**
9. Enter the following data under section [Parameters](#):
 - Under [Business Object Type](#) tab:
[Business Object Type](#): From the dropdown list, choose [Business Partner](#) ([sap.odm.businesspartner.BusinessPartner](#)) with [Business Object Type Version 2.1.1](#).
 - Under API: **MDI_LOG_BUSINESS_PARTNER**
Under [Model Parameters](#) tab:
 - [Mode](#): From the dropdown list, choose [Pull](#)
 - [Package Size](#): **1**
10. Enter the following data under [Descriptions](#):
 - Choose [Create](#).
 - [Language](#): from the dropdown list, choose the relevant language.
 - Repeat the steps for all relevant languages.
11. Enter the following data under [Provider > Provider Interface](#).
 - Choose [Create](#).
Destination: From the dropdown list, choose [MDI](#).
12. Enter the following data under [Consumer](#):
 - Choose [Create](#).
 - [Destination](#): From the dropdown list, select [MDCS](#).
 - [Application Category](#): Do not enter anything.
13. Choose [Save](#) and [Activate](#).

12.4.1.1.6.4 Configure Business System Name and Business Partner SOAP Replication

Procedure

1. Choose your subaccount that has MDI in the subaccount name. Example: CIM_cf-eu12_MDI_Subaccount_2024-4-19-10127.
Log on to this SAP BTP subaccount dedicated for Master Data Integration service.
2. Under [Services](#), choose [Instances and Subscriptions](#) in your subaccount.
3. Under [Instances](#), two Master Data Integration service instances are available. You can choose either one of the two Master Data Integration service instances.

A side view to the right will be opened. Choose [View Credentials](#) and identify the relevant field named [Identityzone](#). The identityzone value is your subaccount name in smallcase.

4. Make a note of this field as you will need to input this value in the [Business System](#) field in the step 8.
5. Under [Subscriptions](#), choose [Master Data Integration \(Orchestration\)](#).
6. Select [Go To Application](#). The system will navigate you to the Fiori launchpad for SAP Master Data Orchestration.
7. Go to section [Configure Master Data Integration](#) and select [Configure Business System Name](#) application.
8. Choose [Create](#), the system will navigate you to the [Configure Business System Name](#) screen. Fill the [Business System](#) field with the identityzone value mentioned in step 3, and choose [Create](#).
9. On the home page, select [Configure SOAP Replication](#) application from the [Configure Master Data Integration](#) section.
In the [Configure SOAP Replication](#) screen, select the [Business System](#). A side view to the right will be opened. Choose [Edit](#). Check the box under [SOAP Business Partner Active](#). Choose [Save](#).

12.4.1.1.6.5 Configure Distribution Model in SAP Business Data Orchestration for Cost Center

Purpose

For data to be replicated between SAP S/4HANA Cloud Public Edition and SAP Master Data Integration, one push model and one pull model need to be created. This chapter introduces how to create a pull replication model for cost center using SAP Business Data Orchestration.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Choose [Master Data Integration \(Orchestration\)](#) from the [Subscriptions](#) section.
4. Choose [Go to Application](#). A new window opens, asking you to select a login option.
5. Choose [Default Identity Provider](#). The Business Data Orchestration home page is displayed.
6. Choose [Manage Distribution Model](#). The [Manage Distribution Model](#) app is displayed.
7. Choose Create.
8. On the next screen, enter the following data under [Header](#):
 - **Model:** [<Your Model Name>](#)

Note

Ensure that the names of replication models in communication arrangements do not include `"/SAP/"` or `"/BDO/"`.

9. Enter the following data under section *Parameters*:
 - Under *Business Object Type* tab:
Business Object Type: From the dropdown list, choose *Cost Center* (*sap.odm.finance.costobject.CostCenter*) with *version 3.0.0*
 - **API: MDI_LOG_COST_CENTER**
 - Under *Model Parameters* tab:
Mode: From the dropdown list, choose *Pull*
Package Size: **1**
10. Enter the following data under *Descriptions*:
 - Choose *Create*.
 - *Language*: From the dropdown list, choose your relevant language.
 - Repeat this step for all relevant languages.
11. Enter the following data under **► Provider ► Provider Interface ►**:
 - Choose *Create*.
 - *Destination*: From the dropdown list, choose *MDI*.
12. Enter the following data under *Consumer*:
 - Choose *Create*.
 - *Destination*: From the dropdown list, select *MDCS*.
 - *Application Category*: Leave empty.
13. Choose *Save* and *Activate*.

12.4.1.1.6.6 Configure Distribution Model in SAP Business Data Orchestration for Project Controlling Object

Purpose

For data to be replicated between SAP S/4HANA Cloud Public Edition and SAP Master Data Integration, one push model and one pull model need to be created. This chapter introduces how to create a pull replication model for project controlling object using SAP Master Data Orchestration.

Procedure

1. Log on to your SAP BTP subaccount dedicated for MDI kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under *Services*, click on *Instances and Subscriptions*.
3. Choose *Master Data Integration (Orchestration)* from the *Subscriptions* section.
4. Choose *Go to Application*. A new window opens, asking you to select a login option.
5. Choose *Default Identity Provider*. The Master Data Orchestration home page is displayed.

6. Choose *Manage Distribution Model*. The *Manage Distribution Model* app is displayed.
7. Choose Create.
8. On the next screen, enter the following data under *Header*:
 - *Model*: **<Your Model Name>**

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

9. Enter the following data under section *Parameters*:
 - Under *Business Object Type* tab:
Business Object Type: From the dropdown list, choose *Project Controlling Object* (*sap.odm.finance.costobject.ProjectControllingObject*) with *version 3.0.0*.
 - API: **MDI_LOG_PROJECT_CONTROLLING_OBJECT**
 - Under *Model Parameters* tab:
 - *Mode*: From the dropdown list, choose *Pull*
 - *Package Size*: **1**
10. Enter the following data under *Descriptions*:
 - Choose *Create*.
 - *Language*: From the dropdown list, choose your relevant language.
 - Repeat this step for all relevant languages.
11. Enter the following data under **Provider > Provider Interface**:
 - Choose *Create*.
 - *Destination*: From the dropdown list, choose *MDI*.
12. Enter the following data under *Consumer*:
 - Choose *Create*.
 - *Destination*: From the dropdown list, select *MDCS*.
 - *Application Category*: Leave empty.
13. Choose *Save* and *Activate*.

12.4.1.1.6.7 Configure Distribution Model in SAP Master Data Orchestration for Exchange Rate

Purpose

To replicate data between SAP S/4HANA Cloud Public Edition and SAP Master Data Integration, you need to create one push model and one pull model. This chapter introduces how to create a pull replication model for exchange rate using SAP Master Data Orchestration.

Procedure

Create Push Model

1. Log on to your SAP BTP subaccount dedicated for kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under *Services*, click on *Instances and Subscriptions*.
3. Choose *Go to Application*. A new window opens, asking you to select a login option.
4. Choose the *Manage Distribution Model* tile. The *Manage Distribution Model* app is displayed.
5. Click *Create*.
6. On the next screen, enter the following data under *Header*:

Model: < Your Model Name >

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

7. Go to *Descriptions*, click *Create* and make the following entries:
 - *Language*: choose the relevant language from your dropdown list.
 - *Item*: Enter description.
8. Make the following entry under tab *Provider* and click *Create*.
Destination: **SAP S/4HANA Cloud Public Edition**
9. Choose *Create*, and enter the following data under tab *Consumer*:
Destination: **MDI**
Application Category: **Nil (Optional)**
10. Under tab *Parameters*, enter the following data:
 - *Business Object Type*: **Exchange Rate (sap.odm.finance.ExchangeRate)**
 - *Business Object Type Version*: after selecting the *Business Object Type*, the version **4.0.0** will be automatically selected from the dropdown list.
 - *Package size*: The number of exchange rates sent along with each request is defined in this field. Choose a number within the range of 1 to 800.
 - *Mode*: **Push**
 - *API*: **MDI_CHANGE_EXCHANGE_RATE**
 - *Recurrence Pattern*: **Daily**
11. Click *Save*. The exchange rate replication model is created, with status as *Inactive*. To activate the replication model, click *Activate* at the top right corner.
12. Click *Replicate*.
In the prompt that appears, choose *Replication Mode: Initial Replication*. Under *Cloud Cockpit Destination*, choose the relevant consuming system.

Create Pull Model

1. Log on to your SAP BTP subaccount dedicated for kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under *Services*, click on *Instances and Subscriptions*.
3. Choose the *Master Data Integration (Orchestration)* tile from the *Subscription* section.
4. Choose *Go to Application*. A new window opens, asking you to select a login option.
5. Choose tile *Manage Distribution Model*. The *Manage Distribution Model* app is displayed.

6. Access the *Manage Distribution Model* app from SAP Business Data Orchestration by going to the subaccount and selecting *Go to Application*.
7. Click *Create*.
8. Under tab *Parameters*, enter the following data:

- *Model*: **Any value**

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

- *Business Object Type*: **Exchange Rate (sap.odm.finance.ExchangeRate)**
 - *Business Object Type Version*: **4.0.0**
 - *Mode*: **Pull**
 - *Package Size*: **10**
 - *Recurrence Pattern*: **Minutes**
 - *API*: **MDI_LOG_EXCHANGE_RATE**
 - *Descriptions*: **optional**
9. Enter the following data under *Descriptions*:
 - Choose *Create*.
 - *Language*: from the dropdown list, choose the relevant language.
 - Repeat the steps for all relevant languages.
 10. Make the following entry under tab *Provider* and click *Create*.
Destination: **MDI**
 11. Enter the following data under tab *Consumer*, and choose *Create*:
Destination: **MDCS**
Application Category: do not enter anything.
 12. Click *Save*. The exchange rate replication model is created, with status as *Inactive*. To activate the replication model, click *Activate* at the top right corner.

Update Service Instance

1. Log on to your SAP BTP subaccount dedicated for kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under *Services*, click on *Instances and Subscriptions*.
3. To create a new instance of the master data integration service, choose *Create*.
In the *New Instance or Subscription* dialog box, make the following entries:
 - *Service*: **Master Data Integration**
 - *Plan*: **sap-integration**
 - *Runtime Environment*: **<Your Model Name>**

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

- *Space*: Select any option from the dropdown list. This value is not relevant for master data integration service.
- *Instance Name*: Enter a service instance name, for example, **<MDI_Service_Instance>**

4. Choose [Next](#).
5. On the next screen, enter the following parameters in JSON format:

Sample Code

```
{ "businessSystemId": "<Business_System_Name>",
  "application": "s4", "writePermissions": [ { "entityType":
"sap.odm.businesspartner.BusinessPartnerRelationship" }, { "entityType":
"sap.odm.businesspartner.BusinessPartner" }, { "entityType":
"sap.odm.businesspartner.ContactPersonRelationship" }, { "entityType":
"sap.odm.finance.costobject.ProjectControllingObject" }, { "entityType":
"sap.odm.finance.costobject.CostCenter" }, { "entityType":
"sap.odm.finance.ExchangeRate" } ] }
```

Note

As service instance already exists, we only update model, and please only update this parameter in the existing documentation.

12.4.1.1.7 Create a Destination inside SAP Ariba Invoicing Subaccount

1. Refer [Retrieve Client Credentials for Master Data Consumption \[page 49\]](#) to collect the following Master Data Integration service key details: clientid, clientsecret and url inside uaa and uri.
2. Log on to your SAP Ariba Invoicing subaccount created in [Create an SAP Subaccount in BTP for SAP Ariba Invoicing \[page 29\]](#).
3. Click [Connectivity->Destinations](#) in the left panel.
4. Click [New Destination](#) and fill the destination with following entries:

Field	Description
Name	Enter a name for this destination.
Type	Select HTTP .
Description	Optional: enter a description for this destination.
URL	Enter the uri collected above.
ProxyType	Select Internet .
Authentication	Select OAuth2ClientCredentials
Client ID	Enter client id collected above.
Client Secret	Enter client secret collected above.
Token Service URL	Enter url collected above.
Token Service URL Type	Choose the default one.

5. In [Additional Properties](#) section, click [New Property](#). Manually enter **sap.mdi.mdcs.hub** in the field with dropdown list, and make the following entries:

Property	Value	Mandatory	Description
sap.mdi.mdc.s.hub	true	Yes	Master Data Consumption Destination

- Click [Save](#).

12.4.1.2 Configuration in SAP S/4HANA Cloud Public Edition

12.4.1.2.1 Create Communication User

Use

In this activity, you create the communication user for outbound communication scenarios.

Procedure

- Log on to the SAP S/4HANA Cloud Public Edition Fiori launchpad with the user that has the role template SAP_BR_ADMINISTRATOR.
- Under the [Communication Management](#) section, navigate to the [Maintain Communication Users](#) application.
- On the [Maintain Communication Users](#) screen, choose [New](#).
- Under [User Data](#), choose one of the following authentication approaches and make the following entries:

Note

It is strongly recommended to use authentication with mTLS. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

- Basic Authentication:**
 - [User Name](#): for example, **4N6_INBOUND**.
 - [Description](#): **<Your description>**.
 - [Password](#): **<Your password>**.
 - Authentication with mTLS**
 - Log on to your subaccount dedicated for the Master Data Integration kernel service and follow the steps in [Create and Download the Public Key of an SAP BTP Managed Certificate \[page 43\]](#). Name the certificate *mdi*.
 - In the [Certificate](#) section, select the certificate you created and click [Upload](#).
- Choose [Create](#).

6. Make a note of user data. This is required when you create the communication system.

12.4.1.2.2 Create Communication System for Supplier Integration (SAP_COM_0008)

Purpose

This procedure describes how to create the communication system that represents the Master Data Integration service instance on BTP for suppliers replication.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an Administrator.
2. Open the [Communication Systems](#) (F1762) app.
3. To create a new communication system, choose [New](#).
The [New Communication System](#) dialog box is displayed.
4. In the [New Communication System](#) dialog box, enter the following data according to your business requirements:

Field Name	Entry Value
------------	-------------

<i>System ID</i>	<Master_Data_Integration_Service>
------------------	-----------------------------------

<i>System Name</i>	<SN_Master_Data_Integration_Service>
--------------------	--------------------------------------

Choose [Create](#).

- On the *Communication System* screen, go to the *General* tab. Enter the following data:

Field Name	Value	Description
Host Name	Note This value is the host of uri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key in Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]: Example uri: https:// one-mds.cfapps.eu10.hana.ondemand.com Host name: one-mds.cfapps.eu10.hana.ondemand.com	Enter the host name of the Master Data Integration service.
Port	443	
Logical System	Enter MDI_BUPA	
Business System	Enter the identityzone from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46] .	

- On the *User for Inbound Communication* tab, choose *Add*. The *New Inbound Communication User* dialog box is displayed.
- In the *New Inbound Communication User* dialog box, enter the following data according to your authentication method:

Note

It is strongly recommended to use authentication method with mTLS certificate. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. Basic Authentication Method

- User Name:** enter the communication user you created in [Create Communication User \[page 17\]](#), for example, **4N6_INBOUND**.

Tip

You can use the [Value Help](#) search for a specific user name.

- Authentication Method:** from the dropdown list, choose *User Name and Password*.

2. Certificate Based Authentication Method

- *Authentication Method*: From the dropdown list, choose *SSL Client Certificate*.
- *User Name/Client ID*: select the user you created in [Create Communication User \[page 59\]](#).

Choose *OK*.

8. On the *User for Outbound Communication* tab, choose *Add*.

The *New Outbound User* dialog box is displayed.

9. In the *New Outbound User* dialog box, enter the following data according to the authentication method you select:

1. Basic Authentication Method

- *Authentication Method*: From the dropdown list, choose *User Name and Password*.
 - *User Name*: Enter the clientid from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#) to get the clientid.
 - *Password*: Enter the clientsecret from the SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#) to get the clientsecret.
- Choose *Create*.

2. mTLS Certificate Based Authentication

- *Authentication Method*: From the dropdown list, choose *OAuth 2.0 mTLS*.
 - *Client Certificate*: Select the *Client Default* certificate.
- Choose *Create*.
- In the *OAuth 2.0 Client Settings*, fill field *mTLS Endpoint* with : certurl + /oauth/token from the service key of [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#).

10. Choose *Save*.

For master data integration, set up communication arrangements as described in the following subchapters.

12.4.1.2.3 Set Up Supplier Integration (SAP_COM_0008)

Use

This procedure describes how to set up the communication scenario SAP_COM_0008 for Business Partner, Customer and Supplier Integration. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an **Administrator**.
2. Open the *Communication Arrangements* (F1763) app.

Note

Ensure that the names of replication models in communication arrangements do not include "/SAP/" or "/BDO/".

- To create a new communication arrangement, choose [New](#).
- In the [New Communication Arrangement](#) dialog box, enter the following data:

Field Name	Entry Value
------------	-------------

<i>Scenario</i>	SAP_COM_0008 (Business Partner, Customer and Supplier Integration)
-----------------	---

→ Tip

You can use the [Value Help](#) to search for a specific scenario.

<i>Arrangement Name</i>	<Arrangement Name>
-------------------------	---------------------------------

- Choose [Create](#).
- On the [Communication Arrangements](#) screen, in the [Common Data](#) section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
<i>Communication System</i>	< Master_Data_Integration_Service >	Select the communication system you created in Create Communication System for Supplier Integration (SAP_COM_0008) [page 60]. After you selected the communication system, the value is also updated for the inbound communication, the inbound services, and the outbound communication.

- On the [Communication Arrangements](#) screen, in the [Outbound Services](#) section, enter the following data for the respective outbound service according to your business requirements:

Outbound Services	Path	Replication Model Example Values	Output Mode	Replication Mode
Business Partner - Replicate from SAP S/4HANA Cloud Public Edition to Client	/ businesspartner/v0/soap/ BusinessPartnerBulkReplicateRequestIn?tenantId=<identityzone>	<BPMDI_CIM> Enter the name for your own replication model.	D (direct output)	C (change replication)

Enter the identityzone from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition](#) [page 46].

→ Tip

If you paste this path, please check whether your pasted path contains spaces. Make sure all spaces are deleted, otherwise supplier replication will not function.

Wrong Path Example: **/businesspartner/v0/ soap/BusinessPartnerBulkReplicateRequestIn? tenantId=**

📌 Note

Activate only the outbound services that are listed in the above table. However, you still need to enter dummy value for some mandatory fields before you can save the communication arrangement.

Note

Once you have done the initial replication of business partners, the next changes to business partners are replicated automatically.

8. Choose [Save](#).

The communication arrangement is now active and all required configurations are generated.

12.4.1.2.4 Set Up Business Partner, Customer and Supplier Blocking Integration (SAP_COM_0091)

Use

This procedure describes how to set up the communication scenario SAP_COM_0091 for business partner, customer and supplier blocking integration. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an Administrator.
2. Open the [Communication Arrangements \(F1763\)](#) app.
3. To create a new communication arrangement, choose [New](#).
4. In the [New Communication Arrangement](#) dialog box, enter the following data:

Field Name	Entry Value
------------	-------------

<i>Scenario</i>	SAP_COM_0091 (Business Partner, Customer and Supplier Blocking Integration)
-----------------	--

→ Tip

You can use the [Value Help](#) to search for a specific scenario.

<i>Arrangement Name</i>	<Arrangement Name>
-------------------------	---------------------------------

5. Choose [Create](#).

- On the [Communication Arrangements](#) screen, in the [Common Data](#) section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
Communication System	< Master_Data_Integration_Service >	Select the communication system you created in Create Communication System for Supplier Integration (SAP_COM_0008) [page 60]. After you selected the communication system, the value is also updated for the inbound communication, the inbound services, and the outbound communication.

- On the [Communication Arrangements](#) screen, in the [Outbound Services](#) section, enter the following data for the respective outbound service according to your business requirements:

Outbound Services	Service Status	Path
ABABusinessPartnerEOPRemoteOut	Active	/businesspartner/v0/soap/BusinessPartnerResponseEOPRemoteIn?tenantId=<identityzone> Enter the identityzone from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]. → Tip If you paste this path, please check whether your pasted path contains spaces. Make sure all spaces are deleted, otherwise supplier replication will not function. E.g.: <i>Wrong Path:</i> / businesspartner/v0/soap/ BusinessPartnerBulk ReplicateRequestIn? tenantId=

Outbound Services	Service Status	Path
ABABusinessPartnerEOPComplout	Active	/businesspartner/v0/soap/BusinessPartnerEOPComplIn?tenantId=<identityzone> Enter the identityzone from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. Please refer to the steps described in chapter Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]. → Tip If you paste this path, please check whether your pasted path contains spaces. Make sure all spaces are deleted, otherwise supplier replication will not function. E.g.: <i>Wrong Path:</i> / businesspartner/v0/soap/ BusinessPartnerBulk ReplicateRequestIn? tenantId=

- Choose [Save](#).
The communication arrangement is now active and all required configurations are generated.

12.4.1.2.5 Create Communication System for SAP Master Data Integration Scenario (SAP_COM_0659)

Purpose

This procedure describes how to create the communication system that represents the Master Data Integration service instance on BTP for replication of cost centers and project controlling objects.

Procedure

- Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an Administrator.
- Open the [Communication Systems](#) (F1762) app.
- To create a new communication system, choose [New](#).
The [New Communication System](#) dialog box is displayed.

- In the *New Communication System* dialog box, enter the following data according to your business requirements:

Field Name **Entry Value**

System ID **<Master_Data_Integration_Service_01>**

System Name **<SN_Master_Data_Integration_Service_01>**

Choose *Create*.

- On the *Communication System* screen, go to the *General* tab. Enter the following data:

Field Name **Value**

Host Name

📘 Note

This value is the host of uri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#):

🔗 Example

uri: <https://one-mds.cfapps.eu10.hana.ondemand.com>

Host name: one-mds.cfapps.eu10.hana.ondemand.com

Port 443

Inbound Only Make sure this checkbox is not checked.

Business System **MDI**

- On the *Communication System* screen, go to the **Technical Data** > *OAuth 2.0 Settings* tab, choose one of the authentication methods and enter following data:

📘 Note

It is strongly recommended to use authentication method with mTLS certificate. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. Basic Authentication Method:

Field Name	Value
Auth. Endpoint	This value is the host of uri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key with a suffix /authorize. For example, <uaa host>/authorize. Note <uaa host> value is the "url" field in the "uaa" section inside service key created in Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]. Example url: https://my-tenant.authentication.eu10.hana.ondemand.com <uaa_host>: my-tenant.authentication.eu10.hana.ondemand.com Auth Endpoint: my-tenant.authentication.eu10.hana.ondemand.com/authorize
Token Endpoint	This value is the host of uri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key with a suffix /oauth/token. For example, <uaa host>/oauth/token. Note <uaa host> value is the "url" field in the "uaa" section inside service key created in Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]. Example url: https://my-tenant.authentication.eu10.hana.ondemand.com <uaa_host>: my-tenant.authentication.eu10.hana.ondemand.com Token Endpoint: my-tenant.authentication.eu10.hana.ondemand.com/oauth/token

2. mTLS Certificate Based Authentication Method:

Field Name	Value
Auth. Endpoint	This value is the host of uri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key with a suffix /authorize. For example, <uaa host>/authorize. Note <uaa host> value is the "url" field in the "uaa" section inside service key created in Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition [page 46]. Example url: https://my-tenant.authentication.eu10.hana.ondemand.com <uaa_host>: my-tenant.authentication.eu10.hana.ondemand.com Auth Endpoint: my-tenant.authentication.eu10.hana.ondemand.com/authorize
mTLS Endpoint	This value is the certuri from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key with a suffix /oauth/token. For example, <certurl>/oauth/token. Example cim-mdi-eu10.authentication.cert.eu10.hana.ondemand.com/oauth/token

7. On the *User for Outbound Communication* tab, choose + (Add). The *New Outbound User* dialog box is displayed.
8. In the *New Outbound User* dialog box, choose your authentication method and enter the following data:
 1. **Basic Authentication Method**
 - *Authentication Method*: From the dropdown list, choose *OAuth 2.0*.
 - *Client Authentication*: *Basic*
 - *OAuth 2.0 Client ID*: Enter the clientid from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. See the steps described in chapter [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#) to get the clientid.
 - *Client Secret*: Enter the clientsecret from SAP Master Data Integration Instance for SAP S/4HANA Cloud Public Edition service key. See the steps described in chapter [Retrieve Client Credentials for SAP S/4HANA Cloud Public Edition \[page 46\]](#) to get the clientsecret.
 2. **mTLS Certificate Based Method**

- [Client Authentication: OAuth 2.0 mTLS](#).
- [Client Secret](#): select the [Client Default](#) certificate.

Choose [Create](#).

9. Choose [Save](#).

For master data integration, set up communication arrangements as described in the following subchapters.

12.4.1.2.6 Create Communication Arrangement (SAP_COM_0659)

Use

This procedure describes how to set up the communication scenario `SAP_COM_0659` for SAP Master Data Integration. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to your SAP S/4HANA Cloud Public Edition system as an Administrator.
2. Open the [Communication Arrangements \(F1763\)](#) app.
3. To create a new communication arrangement, choose [New](#).
4. In the [New Communication Arrangement](#) dialog box, enter the following data:

Field Name	Entry Value
Scenario	SAP_COM_0659 (SAP Master Data Integration)
Arrangement Name	<Your Arrangement Name>

→ Tip

You can use the [Value Help](#) to search for a specific scenario.

5. Choose [Create](#).

- On the [Communication Arrangements](#) screen, in the [Common Data](#) section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
Communication System	<Master_Data_Integration_Service_01>	Select the communication system you created in Create Communication System for SAP Master Data Integration Scenario (SAP_COM_0659) [page 67].

- On the [Communication Arrangements](#) screen, in the [Outbound Communication](#) section, make the following entries:

Field Name	Entry Value
User Name	<Your outbound user created for the communication system>

- On the [Communication Arrangements](#) screen, in the [Outbound Services](#) section, make the following entries:

Field Name	Entry Value
Path	/

- Choose [Save](#).
The communication arrangement is now active and all required configurations are generated.

12.4.1.2.7 Create Communication System for SAP Business Data Orchestration Integration Scenario (SAP_COM_0594)

Purpose

This procedure describes how to create the communication system that represents the Business Data Orchestration service instance on BTP for replication of cost centers and project controlling objects.

Procedure

- Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an **Administrator**.
- Open the [Communication Systems](#) (F1762) app.

- To create a new communication system, choose [New](#).
The [New Communication System](#) dialog box is displayed.
- In the [New Communication System](#) dialog box, enter the following data according to your business requirements:

Field Name	Entry Value
------------	-------------

System ID	<Master_Data_Integration_Service_02>
---------------------------	--------------------------------------

System Name	<SN_Master_Data_Integration_Service_02>
-----------------------------	---

Choose [Create](#).

- In the [General Data](#) section, keep the default system ID and system name.
- In the [Technical Data](#) section, make the following entry:

Field Name	Entry Value
------------	-------------

Inbound Only	Checked
------------------------------	---------

- To add the [User for Inbound Communication](#) area, on the [User for Inbound Communication](#) tab, choose [Add](#).
The [New Inbound Communication User](#) dialog box is displayed.
- In the [New Inbound Communication User](#) dialog box: depending on your authentication methods, choose one of the following options:

Note

It is strongly recommended to use authentication with mTLS. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. Methods other than mTLS Certificate Based Authentication:

- [User Name](#): Enter the communication user you created in [Create Communication User \[page 59\]](#), for example, **4N6_INBOUND**.

→ Tip

You can use the [Value Help](#) to search for a specific user name.

- [Authentication Method](#): From the dropdown list, choose [User Name and Password](#).
Choose [OK](#).

2. Methods with mTLS Certificate Based Authentication:

- [Authentication Method](#): **SSL Client Certificate**
- [User Name/Client ID](#): Select the user created at [Create Communication User \[page 59\]](#).

- Choose [Create](#).
- Choose [Save](#).

12.4.1.2.8 Create Communication Arrangement (SAP_COM_0594)

Use

This chapter describes how to set up the communication scenario **SAP_COM_0594** for SAP Business Data Orchestration. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an **Administrator**.
2. Open the *Communication Arrangement* (F1763) app.
3. To create a new communication arrangement, choose *New*.
4. In the *New Communication Arrangement* dialog box, enter the following data:

Field Name	Entry Value
------------	-------------

<i>Scenario</i>	SAP_COM_0594 (SAP Business Data Orchestration Integration)
-----------------	---

→ Tip

You can use the *Value Help* to search for a specific scenario.

<i>Arrangement Name</i>	<Your Arrangement Name>
-------------------------	--------------------------------------

5. Choose *Create*.
6. On the *Communication Arrangements* screen, in the *Common Data* section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
<i>Communication System</i>	<Master_Data_Integration_Service_02>	Select the communication system you created in <i>Create Communication System for SAP Business Data Orchestration Integration Scenario (SAP_COM_0594)</i> [page 72].

7. Choose *Save*.
The communication arrangement is now active and all required configurations are generated.

12.4.1.3 Set up Destination

12.4.1.3.1 Set up Destinations for SOAP Confirmation

Purpose

A destination needs to be set for SOAP messages for confirmation of Business Partner replication from SAP Master Data Integration to SAP S/4HANA Cloud Public Edition.

Note

Please refer to [Destination Configuration for SAP Master Data Orchestration](#) for the latest procedures to maintain destination for SOAP messages to be sent.

In *Invoice Processing with SAP Ariba Invoicing*, you only need to set one destination BPCONFIRM for SOAP confirmation.

Procedure

1. Log on to your SAP BTP Cockpit subaccount that is dedicated to the kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Go to ► **Connectivity** ► **Destinations** ►.
3. Choose **New Destination** and enter the mandatory details as follows:
 1. Name: <businessSystemId>_BPCONFIRM
The <businessSystemId> value used for this chapter is the same as the one used in [Create Service Instance \[page 42\]](#).
 2. Type: HTTP
 3. URL: The destination URL of the inbound service Business Partner - Receive Confirmation from Client to SAP S/4HANA Cloud Public Edition in the communication arrangement for scenario SAP_COM_0008 created in [Set Up Supplier Integration \(SAP_COM_0008\) \[page 62\]](#).
 4. Proxy Type: Internet
 5. Choose one of the following authentication methods:
 1. **Basic Authentication Method**
 - **Authentication:** **Basic Authentication method**
 - If BasicAuthentication method is selected, and the connected system is SAP S/4HANA Cloud Public Edition, provide username and password for SAP S/4HANA Cloud Public Edition system in [Create Communication System for Supplier Integration \(SAP_COM_0008\) \[page 60\]](#) in the relevant fields in the configuration.
 2. **Client Certificate Authentication**
 - **Authentication:** **Client Certificate Authentication**
 - **Key store location:** certificate created in [Create Service Key \[page 44\]](#).
 - **Key store password:** password for the certificate if there is one.
 - **Use default JDK truststore:** checked

12.4.1.3.2 Set up Destinations for SAP Business Data Orchestration

Purpose

A single destination needs to be set up for each unique client connection.

Procedure

1. Log on to your SAP BTP Cockpit subaccount that is dedicated to the kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Go to ► **Connectivity** ► **Destinations** ► tab and select **New Destination**.
3. Provide the following details:
 1. **Name**: Any name that you would want to identify in the SAP Business Data Orchestration UI.
 2. **Type**: **HTTP**.
 3. **Description**: Provide a description.
 4. **URL**: Provide the service URL of the SAP Business Data Orchestration Distribution Administration service. Look for **MDO: Distribution Administration** among the inbound services that were generated while creating communication arrangement **SAP_COM_0594** ([Create Communication Arrangement \(SAP_COM_0594\) \[page 74\]](#)), and copy the entire URL into this field.
For example: **https://<host>/sap/opu/odata4/sap/mdo_distributionadmin/srvc_a2x/sap/distributionadmin/0001/**
5. **Proxy Type**: **Internet**.
6. **Authentication**:
 1. **Basic Authentication**
User and **Password**: As defined for the inbound communication user assigned to the communication system created in [Create Communication System for SAP Business Data Orchestration Integration Scenario \(SAP_COM_0594\) \[page 72\]](#).
 2. **Client Certificate Authentication**
 - **Authentication**: **Client Certificate Authentication**
 - **Key store location**: certificate created in [Create Service Key \[page 44\]](#).
 - **Key store password**: password for the certificate if one was set
 - **Use default JDK truststore**: checked
4. In the **Additional Properties** section, add the following properties as free text:
 1. **MDOProvider**: **true** (this indicates to SAP Business Data Orchestration that this destination will be used as a possible provider).
 2. **MDOConsumer**: **true** (this indicates to SAP Business Data Orchestration that this destination will be used as a possible consumer).
 3. **MDIInstancelid**: The value of this should be the **systemId** from the service key that you created for the SAP Master Data Integration instance for SAP S/4HANA Cloud Public Edition in Master Data Integration you created in step [Create Service Key \[page 44\]](#).

4. [MDOBusinessSystem](#): The value of this property should be the business system ID of the connected SAP S/4HANA Cloud Public Edition system.
5. Save your entries.

📘 Note

To test if destination connectivity is working fine, please click on [Check Connection](#) button.

12.4.1.4 Set up Distribution Models

Prerequisites

As a prerequisite, a subscription to SAP Business Data Orchestration is required. For details on this, refer to [Subscription Process for SAP Business Data Orchestration](#)

Purpose

Setup distribution models include the destination that was setup for SAP Business Data Orchestration. Based on your scenario, you should create push models. When creating a push model for the same entity type, verify that you create them for the same API version.

For more detailed information, see [Maintenance of the Distribution Model in SAP Business Data Orchestration](#).

📘 Note

The API version should be 3.0.0.

Procedure

Follow the steps below to create a push distribution model wherein data is replicated from an ABAP system to SAP Master Data Integration:

1. Log on to your SAP BTP subaccount dedicated for kernel service created in [Create a Subaccount for SAP Master Data Integration Kernel Service \[page 39\]](#).
2. Under [Services](#), click on [Instances and Subscriptions](#).
3. Choose [Master Data Integration \(Orchestration\)](#) from the [Subscriptions](#) section.
4. Choose [Go to Application](#). A new window opens, asking you to select a login option.
You can now log in with your SAP user and password. If login does not work at the first attempt, please log out and log back in again. The administrator can share the link to the SAP Business Data Orchestration application with appropriate users.
5. Choose [Manage Distribution Model](#) tile. The [Manage Distribution Model](#) app is displayed.
6. Access the [Manage Distribution Model](#) app from SAP Business Data Orchestration by going to the subaccount and selecting [Go to Application](#).

7. Click [Create](#).
8. Specify the following values:
 1. **Model**: Any value (this will determine the name of the DRF model).
 2. **Business Object Type**: select the business object type, such as Cost Center and **Project Controlling Object**.
 3. **Mode**: Push
 4. **Package size**: 1
 5. **Recurrence Pattern**: Pick *continuous distribution (immediate)*. You may also select a distribution pattern if the model's outbound implementation supports change pointers.
 6. **Description**: optional.
 7. **Provider Interface**: Pick from the value help. Select MDI_CHANGE_<business object type> in a push scenario, example:
 - **MDI_LOG_COMPANY_CODE**
 - **MDI_LOG_PROJECT_CONTROLLING_OBJECT**
 8. **Provider**: Pick a provider from the list (ABAP system destination).
 9. **Consumer**: Pick the consumer from the list (SAP Master Data Integration destination).
9. Save your entries.
10. Select this model and click [Activate](#) to activate the model, which will generate the DRF model, and also schedule the below jobs:
 1. A pull job with 5 minutes recurrence to retrieve changed data or confirmations from SAP Master Data Integration, if there is no job scheduled already (e.g. through a pull model).
 2. A job will run to start the initial load from SAP Master Data Integration to the ABAP system, if there is no pull model configured for project controlling object.

📌 Note

WBS push will happen automatically from SAP S/4HANA Cloud Public Edition.

12.5 Invoice, Purchase Order and User Interface Integration

The integration between SAP Ariba Invoicing and SAP S/4HANA Cloud Public Edition is realized via different communication arrangements. Each communication arrangement serves a different function. To realize these functions, communication user, communication system and communication arrangements in SAP S/4HANA Cloud Public Edition must be configured correctly.

1. Invoice and Purchase Order Integration Service
 - Invoices and purchase orders from SAP S/4HANA Cloud Public Edition are replicated to SAP Ariba Invoicing.
 - Invoices and purchase orders received in SAP Ariba Invoicing are posted (or parked) to SAP S/4HANA Cloud Public Edition
2. Invoice user interface integration allows invoice attachments to be displayed in a split screen, as well as a seamless navigation to invoice SAP Fiori apps or SAP Web GUI transactions from connected systems like SAP S/4HANA Cloud Public Edition. This function requires both configuration in SAP S/4HANA Cloud Public Edition and also SAP Ariba Invoicing.

This chapter focuses on the setup steps required in SAP S/4HANA Cloud Public Edition system.

12.5.1 SAP S/4HANA Cloud Public Edition and BTP User Interface Integration

Purpose

The goal of this integration is to integrate the user interface of SAP Ariba Invoicing applications with the SAP S/4HANA Cloud Public Edition Fiori Launchpad. If the same user exists both in SAP S/4HANA Cloud Public Edition and SAP Ariba Invoicing, and is assigned with the same adequate roles, this user can log into SAP S/4HANA Cloud Public Edition Fiori Launchpad and then directly access the relevant SAP Ariba Invoicing applications without any credential requirements or separate login. The end users can perform the necessary actions without noticing the switch of different system landscapes.

In order to achieve this seamless integration, please configure the following communication system and arrangement.

📌 Note

To achieve a seamless integration of SAP BTP and SAP S/4HANA Cloud Public Edition, the following prerequisites must be fulfilled.

1. Create a user using the same email address in both SAP BTP and SAP S/4HANA Cloud Public Edition, and assign the same role templates to this user.
2. Role templates SAP_BR_BPC_EXPERT and SAP_BR_AP_ACCOUNTANT should be assigned to your user in the SAP S/4HANA Cloud Public Edition landscape. SAP_BR_AP_ACCOUNTANT allows you to access app [Manage Supplier Invoices Centrally](#) and [Upload Supplier Invoices Centrally](#) under tab [Invoicing](#). SAP_BR_BPC_EXPERT allows you to access apps under tab [Central Invoice Administration](#). The same user using the same email address should also exist in SAP BTP.

12.5.1.1 Communication Setup in SAP S/4HANA Cloud Public Edition

12.5.1.1.1 Configure the Connected System - SAP S/4HANA Cloud Public Edition

12.5.1.1.1.1 Create Communication User

Use

In this activity, you create the communication user for the outbound communication scenarios.

Procedure

1. Log on to the SAP S/4HANA Cloud Public Edition Fiori UI launchpad with the user that has the role template `SAP_BR_ADMINISTRATOR`.
2. Under the *Communication Management* section, navigate to the *Maintain Communication Users* application.
3. On the *Maintain Communication Users* screen, choose *New*.
4. Under *User Data*, choose one of the following authentication approaches and make the following entries:

Note

It is strongly recommended to use authentication with mTLS. Basic authentication method is only for compatibility purpose, and will be obsolete in future.

1. **Basic Authentication:**
User Name: for example, **4N6_INVOICES**.
Description: **Your description**.
Password: **Your password**.
2. **Authentication with mTLS**
Log on to your subaccount dedicated for the SAP Invoicing kernel service and follow the steps in [Create and Download the Public Key of an SAP BTP Managed Certificate \[page 43\]](#). Name the certificate *cim*.
In the *Certificate* section, select the certificate you created and click *Upload*.
5. Choose *Create*.
6. Make a note of user data. This is required when you create the communication system.

12.5.1.1.1.2 Create Communication System for Invoice and Purchase Order Integration

Use

In this activity, you create the communication system for communication arrangement SAP_COM_0516 for connected system. This enables invoice and purchase order data to be exchanged between connected system and the SAP BTP Hub system.

Procedure

1. Log on to the SAP S/4HANA Cloud Public Edition Fiori UI launchpad with the user that has the role template SAP_BR_ADMINISTRATOR.
2. In the *Communication Management* section, navigate to the *Communication Systems* (F1762) application.
3. To create a new communication system, on the *Communication System* screen, choose *New*. The *New Communication System* dialog box is displayed.
4. On the *New Communication System* screen, enter the following entries and choose *Create*.

Field Name	Entry Value
------------	-------------

<i>System ID</i>	<SAP_Central_Invoice_Management>
------------------	----------------------------------

ⓘ Note

The length of system ID should not exceed 10 digits.

<i>System Name</i>	<SAP_Central_Invoice_Management_System_Name>
--------------------	--

ⓘ Note

Make a note of this name as it is required in later steps.

5. In the *General Data* section, keep the default system ID and system name.
6. In the *Technical Data* section, make the following entry:

Field Name	Entry Value
------------	-------------

<i>Inbound Only</i>	Checked
---------------------	---------

7. To add the *User for Inbound Communication* area, on the *User for Inbound Communication* tab, choose *Add*. The *New Inbound Communication User* dialog box is displayed.

8. In the *New Inbound Communication User* dialog box, enter the following data:
 - *User Name*: Enter the communication user you created in [Create Communication User \[page 80\]](#), for example, **4N6_INVOICES**.

→ Tip

You can use the [Value Help](#) to search for a specific user name.

- *Authentication Method*: From the dropdown list, choose *User Name and Password*.
- Choose *OK*.

9. Choose *Create*.
10. Choose *Save*.

12.5.1.1.1.3 Set Up Central Invoice and Purchase Order Satellite Integration (SAP_COM_0516)

Use

In this activity, you create a communication arrangement that provides the required metadata for the service configuration, such as credentials, outbound ports, destinations, URLs and other service specifications for invoice and purchase order data replication.

Procedure

1. Log on to the SAP Fiori launchpad of your SAP S/4HANA Cloud Public Edition system as an **Administrator**.
2. In the *Communication Management* section, navigate to the *Communication Arrangements* (F1763) application.
3. On the *Maintain Communication Arrangements* screen, choose *New*.
4. On the *New Communication Arrangement* screen, make the following entries and then choose *Create*:

Field Name	Entry Value
<i>Scenario</i>	SAP_COM_0516 (SAP S/4HANA Cloud Public Edition Procurement Hub - Central Invoice Satellite Integration)

→ Tip

You can use the [Value Help](#) to search for a specific scenario.

Field Name	Entry Value
------------	-------------

<i>Arrangement Name</i>	<Your Communication Arrangement Name>
-------------------------	---------------------------------------

- On the *Communication Arrangements* screen, in the *Common Data* section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
<i>Communication System</i>	Select the communication system you created in Create Communication System for Invoice and Purchase Order Integration [page 81].	

- Choose *Save*.
The communication arrangement is now active and all required configurations are generated.

12.5.1.1.2 Create Communication System for User Interface Integration

Use

In this activity, you create the communication system for user interface integration.

Procedure

- Log on to the SAP S/4HANA Cloud Public Edition Fiori UI launchpad with the user that has the role template SAP_BR_ADMINISTRATOR.
- In the *Communication Management* section, navigate to the *Communication Systems* (F1762) application.
- To create a new communication system, choose *New* on the *Communication Systems* screen.
The *New Communication System* dialog box is displayed.
- On the *New Communication System* screen, enter the following entries and choose *Create*.

Field Name	Entry Value
------------	-------------

<i>System ID</i>	Your system ID, for example, <COM_SAP_0897>
------------------	--

Note

The length of system ID should not exceed 10 digits.

<i>System Name</i>	Your system name, <COM_SAP_0897>
--------------------	---

Note

Note down this name as it is required in later steps.

5. In the *General Data* section, make the following entries:

Field Name	Entry Value
------------	-------------

<i>Host Name</i>	<Your BTP Tenant Host URL>
------------------	---

Note

For the tenant host URL, please enter your SAP Ariba Invoicing tenant host.

For example: **cim-tes-eu10-dev-exp-test123.eu10.cim.cloud.sap**

<i>Port</i>	<Default Port Value>
-------------	-----------------------------------

6. Choose *Save*.

12.5.1.1.3 SAP Ariba Invoicing - User Interface Integration(SAP_COM_0897)

Use

This procedure describes how to set up the communication scenario SAP_COM_0897 for SAP Ariba Invoicing - User Interface Integration. For the communication scenario, a communication arrangement is needed.

Procedure

1. Log on to the SAP Fiori Launchpad with user that has the role template SAP_BR_ADMINISTRATOR .

2. In the *Communication Management* section, navigate to the *Communication Arrangements* (F1763) application.
3. To create a new communication arrangement, choose *New*.
4. In the *New Communication Arrangement* dialog box, make the following entries and then choose *Create*:

Field Name	Entry Value
Scenario	SAP_COM_0897 (SAP Ariba Invoicing - User Interface Integration)
Arrangement Name	<Your Communication Arrangement Name>
Communication System	<Your communication system>

→ Tip

You can use the *Value Help* to search for a specific scenario.

5. On the *Communication Arrangements* screen, in the *Common Data* section, enter the following data according to your business requirements:

Field Name	Entry Value	Description
Communication System	<Your communication system>	Select the communication system you created in Create Communication System for User Interface Integration [page 83].

6. In the **Outbound Services** > *Navigation to CIM BTP* section, you can decide to check the *User Embedded* box or not.

ⓘ Note

If you check *Use Embedded*, then BTP applications will open in the same window. If you don't check *Use Embedded*, then BTP applications will open in different windows.

ⓘ Note

If the *Use Embedded* option is checked in the communication arrangement, users must add an exception for the domain of the identity authentication tenant to ensure that single sign-on is working. Some examples of domains are **<xxx>.accounts.cloud.sap** and **xxx.accounts.ondemand.com**. Users can consult the documentation of the respective browsers for more information about how to enable third-party cookies for specific websites and domains.

7. Choose *Save*.
The communication arrangement is now active and all required configurations are generated.

12.6 Configure SAP Ariba Invoicing on SAP Business Technology Platform (BTP)

Please carry out the following steps in this chapter.

12.6.1 Create a Destination for an SAP S/4HANA Cloud Public Edition System

You need to create one or more destinations that point to the system to which you want to connect.

Context

You need to create only one destination. A single destination can cover all scenarios at the same time, as all scenarios are covered by a single communication arrangement related to the **SAP_COM_0516** scenario. This communication arrangement is defined in your connected system.

Procedure

1. Log on to your SAP BTP Cockpit subaccount that is dedicated to SAP Ariba Invoicing created in [Create an SAP Subaccount in BTP for SAP Ariba Invoicing \[page 29\]](#).
2. In the left menu, click ► **Connectivity** ► **Destinations** ►.
3. Click **New Destination** and fill the destination with the following content:

Field	Description
<i>Name</i>	Enter a name for this destination. Example: CONNECTED_SYSTEM_<your business system ID>
<i>Type</i>	Select HTTP .
<i>Description</i> (Optional)	Enter a description for this destination.

Field	Description
URL	Enter the API-URL from the communication arrangement related to the SAP_COM_0516 scenario defined in your connected system. Example: https://<mytenant>-api.s4hana.ondemand.com Note When the OData service is used, the destination will be automatically completed with the communication arrangement path. Example for replication for supplier invoices: /sap/opu/odata/sap/CIMV2_SI_BULK_EXTRACT_SRV/ will be added after the destination URL. See About Communication Arrangements for more information.
ProxyType	Select Internet .
Authentication	Select ClientCertificateAuthentication or BasicAuthentication. Note We recommend to use ClientCertificateAuthentication. To do so, you need to upload your own certificate to the key store: 1. Generate a certificate. 2. Click Certificates above the list. 3. Upload your certificate by clicking Upload Certificate. Only .pfx, .p12 and .jks files are allowed.
User/Password or ClientCertificateAuthentication	• If you have selected BasicAuthentication, use the name and password of the inbound communication user from the communication arrangement related to the SAP_COM_0516 scenario. This communication arrangement is defined in your connected system. • If you have selected ClientCertificateAuthentication, select your certificate in the dropdown list.

4. In the [Additional Properties](#) section, click [New Property](#) and create the following property:
 - Enter [sap.procure.hub](#) for the property name, and enter **true** for the value. This property is required to make the destination visible in SAP Ariba Invoicing.
5. Click [Save](#).

Related Information

Create HTTP Destinations

12.6.2 Subscribe to Inbound Processing Module

Inbound processing module is an additional module for SAP Ariba Invoicing that allows users to upload invoice files, edit draft invoices and save supplier invoices to a connected system, directly from SAP Ariba Invoicing apps.

📘 Note

For information about current restrictions related to SAP Ariba Invoicing, inbound processing module, please read SAP Note [3331880](#) sections **2. Supported Languages and Countries/Regions** and **3. Restrictions for SAP Invoicing, Inbound Processing Module**.

Subscribing to SAP Ariba Invoicing, inbound processing module

Prerequisites

- You have already configured SAP Ariba Invoicing and you have an active SAP Ariba Invoicing subscription.
- In SAP BTP cockpit, in the [Entitlements](#) page of your subaccount, you have added [SAP Ariba Invoicing, inbound processing module](#) with [standard \(Application\)](#) plan. See [Managing Entitlements and Quotas Using the Cockpit](#).

Subscription Procedure

In SAP BTP cockpit, navigate to the [Service Marketplace](#), select the [SAP Ariba Invoicing, inbound processing module](#) tile and click [Create](#). In the [New Instance or Subscription](#) window:

- For service, select [SAP Ariba Invoicing, inbound processing module](#).
- For plan, leave it on [standard](#), no matter if you're configuring a test tenant or a productive tenant.
The inbound processing module will be activated in the [SAP Ariba Invoicing](#) tenant created in this subaccount, so the inbound processing module will use the same plan as the existing [SAP Ariba Invoicing](#) subscription. Even if you are configuring a test tenant, you have to leave the subscription plan for [SAP Ariba Invoicing, inbound processing module](#) on [standard](#).

To create a	For SAP Ariba Invoicing subscription plan, select	For SAP Ariba Invoicing, inbound processing module subscription plan, select
Test tenant	test	standard
Productive tenant	standard	standard

📘 Note

Once the subscription is complete, an application named [SAP Ariba Invoicing, Inbound Processing Module](#) is automatically created in the Identity Authentication service (IAS) tenant. Please do not remove this application while your subscription is active.

📘 Note

After you have subscribed, please note that there may be a 20 minute delay before new features are available in SAP Ariba Invoicing apps.

12.7 Retrieve Value for Target System Connection

Context

As part of the Booster setup, an instance of service **SAP Ariba Invoicing** with plan **scim** was created in your subaccount dedicated to SAP Ariba Invoicing.

Procedure

1. Navigate to your subaccount dedicated to SAP Ariba Invoicing.
2. Go to *Instances and Subscriptions*, under *Instances*, look for an instance with the service name **SAP Ariba Invoicing**.
3. Click on the instance. Under **Service bindings**, you will find a service binding with the name **cim-scim-api-key**.
4. Open this service binding and note down the values of the following fields. You will need them for the target system connection setup.
 - scim-v2
 - clientid
 - clientsecret
 - url

12.8 Set up Target System for User Replication

Context

User replication allows you to copy user information like username, email and other fields like given name or family name, depending on the import configuration from a source system to your SAP Ariba Invoicing tenant (target system). A source system can be, for example, an **SAP Cloud Identity Service – Identity Authentication** tenant.

You must configure user replication to enable the following:

- Display user information in various SAP Ariba Invoicing apps. The information include the users that are editing a draft invoice, participating in an invoice workflow, or have uploaded an invoice file.
- Personal data reporting for users using the **SAP Personal Data Manager** service. See [Personal Data Reporting](#) for more information.
- Blocking and deleting personal data of users using the **SAP Data Retention Manager** service. See [Personal Data Deletion](#) for more information.

Note

User replication does not include existing role assignments from the connected system. SAP Ariba Invoicing roles are independently managed in the SAP BTP cockpit. Roles give administrators or users access to SAP Ariba Invoicing apps, and are assigned later during the configuration procedure. Once a user logs on to the SAP Ariba Invoicing launchpad, the email from their session will be used to match the user information replicated to SAP Ariba Invoicing.

User information may be displayed in SAP Ariba Invoicing although the user does not have, or need access to SAP Ariba Invoicing apps. This could happen, for instance, for a user participating in an invoice workflow if workflow extraction is enabled.

You can use Identity Provisioning to configure SAP Ariba Invoicing as a target system where you can provision users and groups from source systems.

Procedure

1.
 - **If you use SAP Cloud Identity Services, follow the steps below:**
Sign in to the administration console of SAP Cloud Identity Services and navigate to ► [Identity Provisioning](#) ► [Target Systems](#) ►.
 - **If you use Identity Provisioning Service in Neo Environment, access the Identity Provisioning UI by:**
 - [Access Identity Provisioning UI of Bundle Tenants](#)
 - [Access Identity Provisioning UI of Standalone Tenants](#)
2. Add [SCIM System](#) as a target system. For more information, see [Add New Systems](#).
3. Choose the [Properties](#) tab to configure the connection settings for your system.

Note

If your tenant is running on SAP BTP, Neo environment, you can create a connectivity destination in your subaccount in the SAP BTP cockpit by following the instructions in [Create Destinations \(Cockpit\)](#). Then, select this destination from the [Destination Name](#) combo box in your Identity Provisioning screen.

If one and the same property exists both in the cockpit and in the [Properties](#) tab, the value set in the [Properties](#) tab is considered with higher priority.

We recommend that you use the [Properties](#) tab. Use a connectivity destination only if you need to reuse one and the same configuration for multiple provisioning systems.

Use the value you noted down in [Retrieve Value for Target System Connection \[page 89\]](#).

Table 3: Mandatory Properties

Property Name	Value
<i>Type</i>	HTTP
<i>URL</i>	Use the <i>scim-v2</i> value you noted down in step 4 of Retrieve Value for Target System Connection [page 89], and remove the path information. Example: https://eu10.cim.cloud.sap
<i>ProxyType</i>	<i>Internet</i>
<i>Authentication</i>	<i>BasicAuthentication</i>
<i>User</i>	Enter the <i>clientid</i> value you noted down in step 4 of Retrieve Value for Target System Connection [page 89] .
<i>Password</i>	(Credential) Enter the <i>clientsecret</i> value you noted down in step 4 of Retrieve Value for Target System Connection [page 89] .
<i>OAuth2TokenServiceURL</i>	Enter the <i>url</i> value you noted down in step 4 of Retrieve Value for Target System Connection [page 89], and add /oauth/token to the end. Example: https://<my tenant>.authentication.eu10.hana.ondemand.com/oauth/token
<i>cim.user.unique.attribute</i> (Optional)	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Default value: <i>userName</i> If the service finds an existing user by <i>userName</i>, it updates this user with the data of the conflicting one. If the service does not find an existing user by <i>userName</i>, the creation of the conflicting user fails.

Property Name	Value
<i>cim.group.prefix</i> (Optional)	This property distinguishes SAP Ariba Invoicing groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: CIM_ You can use the example value or provide your own. When set in the target system, only groups containing the CIM_ prefix in their display name will be provisioned to SAP Ariba Invoicing. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Ariba Invoicing.
<i>ips.delete.threshold.groups</i> (Optional)	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties
<i>ips.delete.threshold.users</i> (Optional)	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties

To learn what additional properties are relevant to this system, see [List of Properties](#). You can use the main search, or filter properties by the [Name](#) or [System Type](#) columns.

4. Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the [SAP Ariba Invoicing](#) target system, whose settings are displayed under the [Transformations](#) tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Invoicing system. For more information, see [Manage Transformations](#).

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Ariba Invoicing entity.

Default transformation:

Code Syntax

```
{ "user": { "mappings":
[ { "constant": [ "urn:ietf:params:scim:schemas:core:2.0:User" ],
"preserveArrayWithSingleElement": true, "targetPath": "$.schemas" },
{ "sourceVariable": "entityIdTargetSystem", "targetPath": "$.id" },
{ "sourcePath": "$.userName", "targetPath": "$.userName" }, { "sourcePath":
"$.externalId", "targetPath": "$.externalId", "optional": true },
{ "sourcePath": "$.name.givenName", "targetPath": "$.name.givenName",
"optional": true }, { "sourcePath": "$.name.middleName", "targetPath":
```

```

"$name.middleName", "optional": true }, { "sourcePath": "$name.familyName",
"targetPath": "$name.familyName", "optional": true }, { "sourcePath":
"$name.formatted", "targetPath": "$name.formatted", "optional":
true }, { "sourcePath": "$name.honorificPrefix", "targetPath":
"$name.honorificPrefix", "optional": true }, { "sourcePath":
"$name.honorificSuffix", "targetPath": "$name.honorificSuffix", "optional":
true }, { "sourcePath": "$emails", "preserveArrayWithSingleElement":
true, "optional": true, "targetPath": "$emails" }, { "sourcePath":
"$active", "targetPath": "$active", "optional": true }, { "sourcePath":
"$displayName", "targetPath": "$displayName", "optional": true },
{ "sourcePath": "$nickName", "targetPath": "$nickName", "optional":
true }, { "sourcePath": "$profileUrl", "targetPath": "$profileUrl",
"optional": true }, { "sourcePath": "$title", "targetPath": "$title",
"optional": true }, { "sourcePath": "$userType", "targetPath":
"$userType", "optional": true }, { "sourcePath": "$preferredLanguage",
"targetPath": "$preferredLanguage", "optional": true }, { "sourcePath":
"$locale", "targetPath": "$locale", "optional": true }, { "sourcePath":
"$timezone", "targetPath": "$timezone", "optional": true } ] },
"group": { "condition": "('%cim.group.prefix%' === 'null') ||
($displayName =~ /%cim.group.prefix%.*/)", "mappings": [ { "sourceVariable":
"entityIdTargetSystem", "targetPath": "$id" }, { "constant": [
"urn:ietf:params:scim:schemas:core:2.0:Group" ], "targetPath": "$schemas" },
{ "sourcePath": "$externalId", "targetPath": "$externalId", "optional":
true }, { "sourcePath": "$displayName", "targetPath": "$displayName",
"functions": [ { "condition": "('%cim.group.prefix%' !== 'null') &&
(@ =~ /%cim.group.prefix%.*/)", "function": "replaceFirstString",
"regex": "%cim.group.prefix%", "replacement": "" } ] }, { "sourcePath":
"$members[*].value", "preserveArrayWithSingleElement": true, "optional":
true, "targetPath": "$members[?(@.value)]", "functions": [ { "function":
"resolveEntityIds" } ] } ] } ] } } }

```

5. Add a source system from which to read users. The two most frequently used source systems are [Identity Authentication](#) and [SAP S/4HANA Cloud Public Edition](#). You can follow the steps below to create your source systems if there is none available.
 - [Source system for Identity Authentication](#)
 - [Source system for SAP S/4HANA Cloud Public Edition](#)
 - For other source systems, see [Source System](#).

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you used in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see [Manage Job Notifications](#).
- Now, start an identity provisioning job. For more information, see [Monitor Provisioning Job Logs](#).

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